Bioinformatics



KB8029

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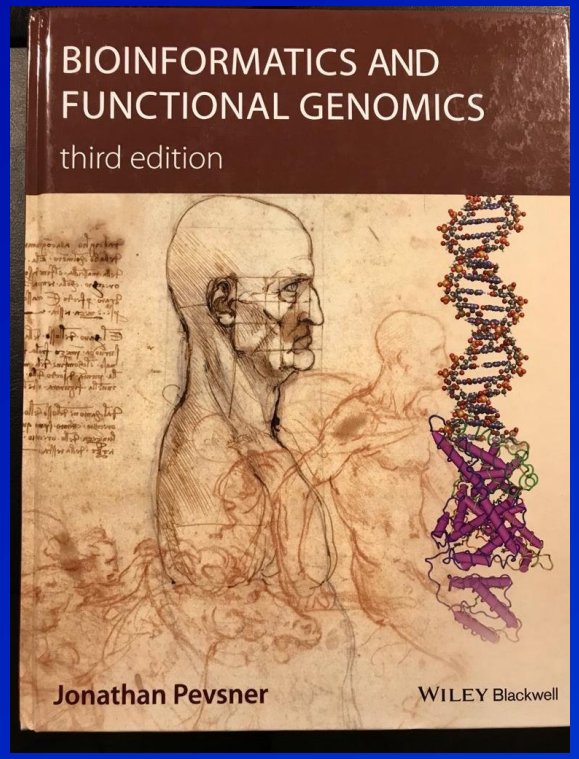
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Advanced Database Search



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Basic Local Alignment Search Tool (BLAST) CHAPTER 4

Advanced Database Searching CHAPTER 5

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Advanced Database Search

Word search (BLAST)

Profile search (PSI-BLAST)

Profile Hidden Markov Model

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Heuristic Methods

Complexity in searching large databases can be overcomed by:

- **Implementation dynamic programming in hardware**
- **Parallelization of the search**
- **Using approximate methods (heuristics)**

Most common heurictic methods

based on observations that:

- ***Substitutions are much more likely than insertions/deletions***
- ***Good alignments contain many matched substrings without gaps***

require that:

- ***database is preprocessed (hash table)***

and use

- ***matched substrings as starting points for further search***

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BLAST

Basic Local Alignment Search Tool

S.F. Altschul *et al.*, J.Mol.Biol (1990): 215, 493-410; Nucl. Acid Res (1997): 25, 3389-3402

Algorithm



Compiles list of word pairs from the query sequence

Scans database for hits and extends them.

Evaluates statistical significance of produced alignments

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Popularity of BLAST

- **Historical dominance and legacy**
  - Integration into many scientific workflows
- **Fast and efficient algorithm**
  - Optimized for search in large databases
  - Sufficient for the most routine sequence comparisons
- **Broad range of applications**
  - Flexibility in sequence and alignment types
- **Widely supported**
  - Continuous development and updates
  - Large user database

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BLAST Algorithm

Stage I. Seeding

Precomputed scores for alignments of short protein fragments (words)

$n=3$

$N=20^3=8000$

	AAA	…	LWG	…	YYY
AAA	12	…	-4	…	-6
…	…	…	…	…	…
IWG	-4	…	19	…	-2
…	…	…	…	…	…
LWA	0	…	15	…	-1
…	…	…	…	…	…
YYY	-6	…	-2	…	-21

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BLAST Algorithm

Stage I. Seeding

Precomputed scores for alignments of short protein fragments (words)

$n=3$

$N=20^3$

$=8000$

	1	…	2452	…	8000
1	12	…	-4	…	-6
…	…	…	…	…	…
2152	-4	…	19	…	-2
…	…	…	…	…	…
3655	0	…	15	…	-1
…	…	…	…	…	…
8000	-6	…	-2	…	-21

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BLAST Algorithm

Stage I. Seeding

Precomputed scores for alignments of short protein fragments (words)

$n=3$

$N=20^3=8000$

	AAA	…	LWG	…	YYY
AAA	12	…	-4	…	-6
…	…	…	…	…	…
IWG	-4	…	19	…	-2
…	…	…	…	…	…
LWA	0	…	15	…	-1
…	…	…	…	…	…
YYY	-6	…	-2	…	-21

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BLAST Algorithm

Stage I. Seeding

Precomputed scores for alignments of short protein fragments (words)

Scores with BLOSUM62

S(LWG,IWG)= S(L,I)+ S(W,W)+ S(G,G)

S(LWG,IWG)= 2 + 11 + 6 = 19

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4

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BLAST Algorithm

Stage I. Seeding

Precomputed scores for alignments of short protein fragments (words)

Scores with BLOSUM62

S(LWG,IWG)= S(L,I)+ S(W,W)+ S(G,G)
S(LWG,IWG)= 2 + 11 + 6 = 19

S(AAA,IWG)= S(A,I)+ S(A,W)+ S(A,G)
S(LWG,IWG)= -1 + -3 + 0 = -4

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	-1	1
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4

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BLAST Algorithm

Stage I. Seeding

Compile list of words

Query sequence

…VTALWGKNVD…

3-letters words derived from the query sequence

… VTA TAL ALW LWG WGK GKN KNV NVD …

Sort a column of the precomputed table corresponding to a word from the query sequence in descending order and extract rows with values above a threshold T (*usually* 12 *for proteins*),

	LWG
LWG	21
IWG	19
MWG	19
VWG	18
FWG	17
AWG	17
LWS	15
LWN	15
LWA	15
LYG	12
LPG	11
…	…

← Threshold

Generate similar lists for all words

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BLAST Algorithm

Stage II. Extension

Pre-calculate positions of words in the sequences of the databases

Database sequence LSPADKTNVKAAWGKVGAHAGESFEALERMFLSFSHTKTYFP…

Hash table

Word	Position(s)
AAA	0
…	…
AWG	12
…	…
PAD	3
…	…
SFP	23, 34
…	…
YYY	0

No such word in the sequence

Number or hash indices

Position(s) of the word start in the sequence

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BLAST Algorithm

Stage II. Extension

Find hits in the database and extend them

Query sequence LTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYWTQRFFESFGDLSTPDAVMGNPKV

Database sequence LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQV

HSP

LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPKV
LSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHF------DLSHGSAQV

WG
AWG

Extension

Words triggering extension

Extension

Where to stop extension?

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BLAST Algorithm

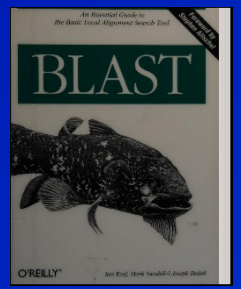
Stage II. Extension

Where to stop extension?

Query sequence	`The quick brown fox jumps over the lazy dog.`
Database sequence	`The quiet brown cat purrs when she sees him.`
Hit	`The`
Scoring	`+1 for exact match, -1 otherwise, no gaps`
Drop-off score threshold	`5`
Extension	

Words triggering extension

The
The



I. Korf, M. Yandell, and J. Bedell. BLAST. O'Reilly Media (2003), pp. 79 - 87

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BLAST Algorithm

Stage II. Extension

Where to stop extension?

Query sequence `The quick brown fox jumps over the lazy dog.`

Database sequence `The quiet brown cat purrs when she sees him.`

Hit `The`

Scoring `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold `5`

Extension

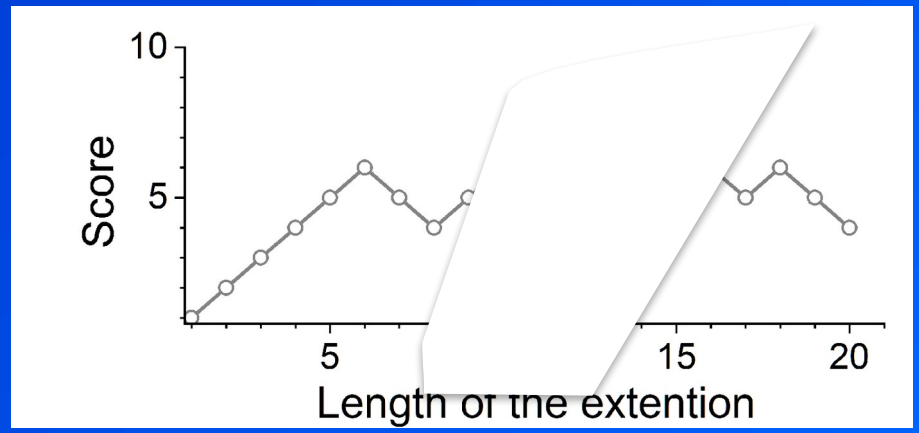
```
The qui
The qui
123 456
000 000
```

Words triggering extension

Raw score

Drop-off score





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BLAST Algorithm

Stage II. Extension

Where to stop extension?

Query sequence `The quick brown fox jumps over the lazy dog.`

Database sequence `The quiet brown cat purrs when she sees him.`

Hit `The`

Scoring `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold `5`

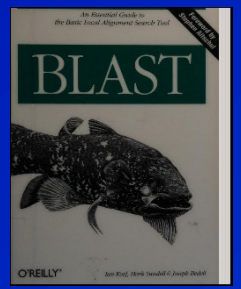
Extension

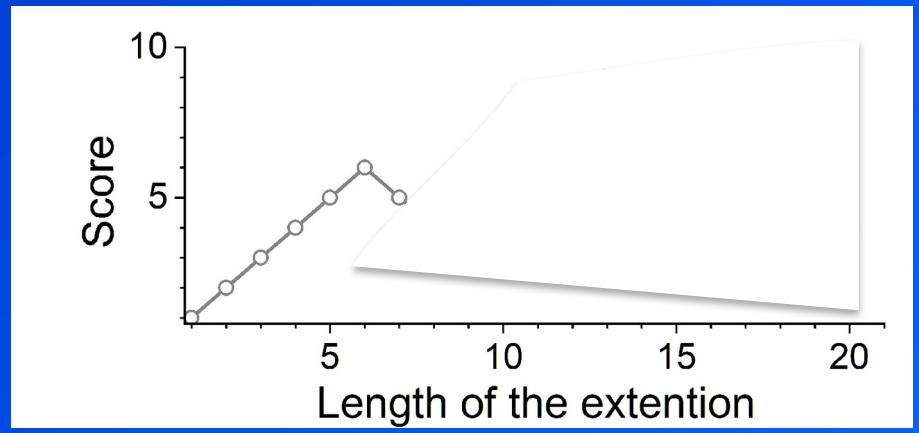
```
The quic
The quie
123 4565
000 0001
```

Words triggering extension

Raw score

Drop-off score





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BLAST Algorithm

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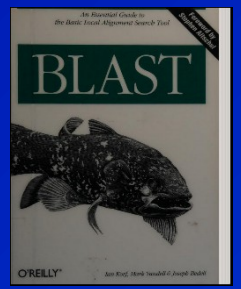
Extension

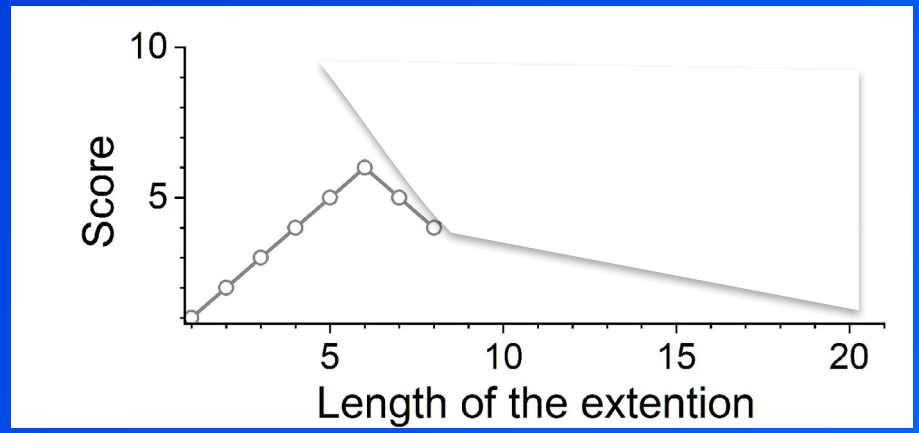
```
The quick
The quiet
123 45654
000 00012
```

Words triggering extension

Raw score

Drop-off score





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BLAST Algorithm

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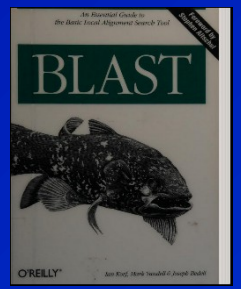
Extension

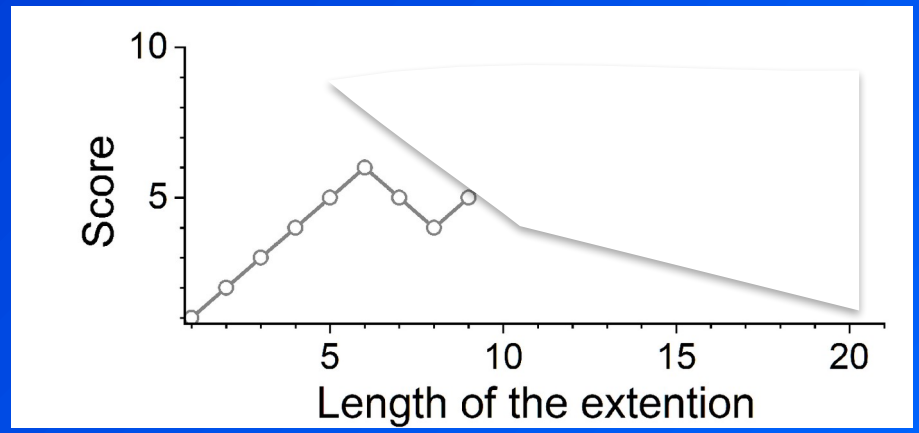
```
The quick b
The quiet b
123 45654 5
000 00012 1
```

Words triggering extension

Raw score

Drop-off score





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BLAST Algorithm

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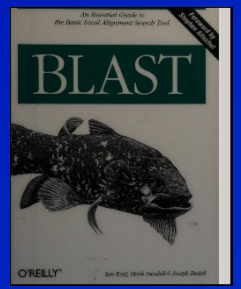
Extension

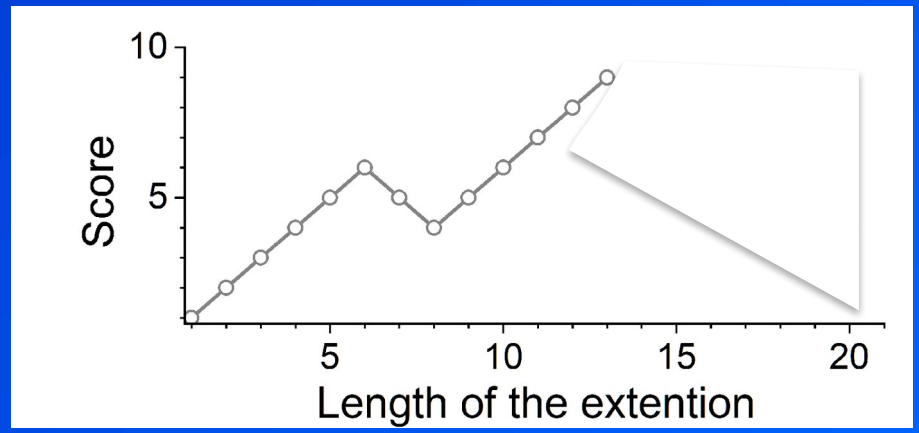
```
The quick brown
The quiet brown
123 45654 56789
000 00012 10000
```

Words triggering extension

Raw score

Drop-off score





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Hit `The`

Scoring `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold `5`

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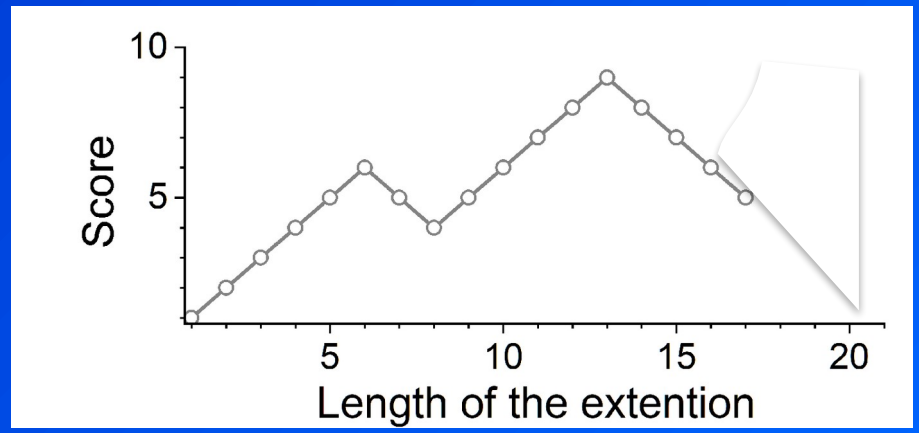
```
The quick brown fox j
The quiet brown cat p
123 45654 56789 876 5
000 00012 10000 123 4
```

Words triggering extension

Raw score

Drop-off score





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BLAST Algorithm

Stage II. Extension

Where to stop extension?

Query sequence — `The quick brown fox jumps over the lazy dog.`

Database sequence — `The quiet brown cat purrs when she sees him.`

Hit — `The`

Scoring — `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold — `5`

Extension

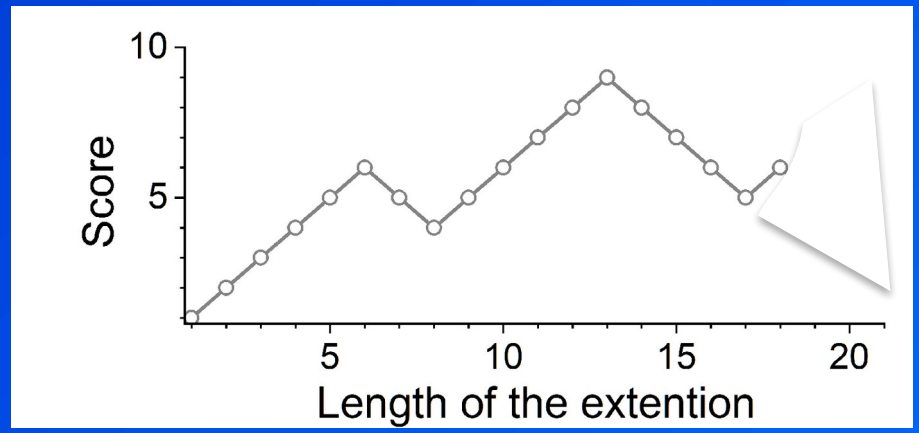
```
The quick brown fox ju
The quiet brown cat pu
123 45654 56789 876 56
000 00012 10000 123 43
```

Words triggering extension

Raw score

Drop-off score





I. Korf, M. Yandell, and J. Bedell. BLAST. O'Reilly Media (2003), pp. 79 - 87

Stockholms universitet

BLAST Algorithm

Stage II. Extension

Where to stop extension?

Query sequence — `The quick brown fox jumps over the lazy dog.`

Database sequence — `The quiet brown cat purrs when she sees him.`

Hit — `The`

Scoring — `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold — `X=5`

Extension

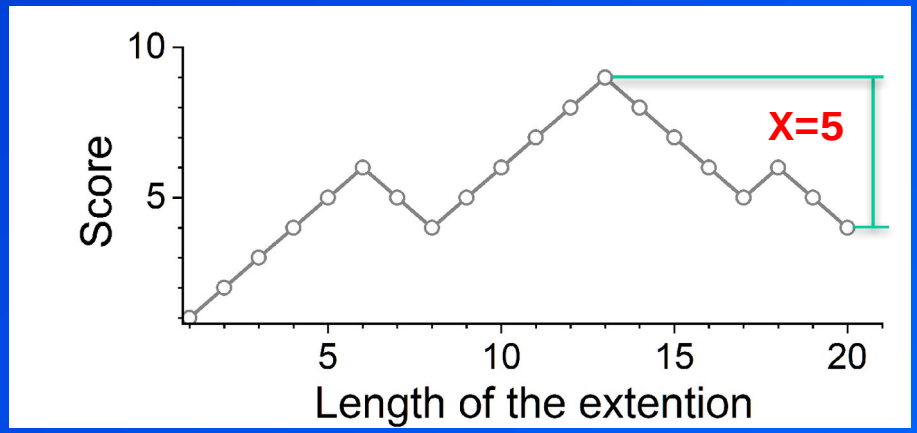
```
The quick brown fox jump
The quiet brown cat purr
123 45654 56789 876 5654
000 00012 10000 123 4345
```

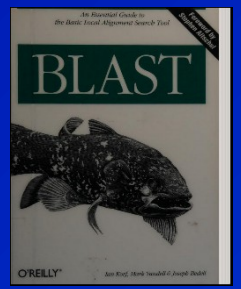
Words triggering extension

Raw score

Drop-off score

Threshold achieved





I. Korf, M. Yandell, and J. Bedell. BLAST. O'Reilly Media (2003), pp. 79 - 87

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BLAST Algorithm

Stage II. Extension

Where to stop extension?

Query sequence — `The quick brown fox jumps over the lazy dog.`

Database sequence — `The quiet brown cat purrs when she sees him.`

Hit — `The`

Scoring — `+1 for exact match, -1 otherwise, no gaps`

Drop-off score threshold — `5`

Extension

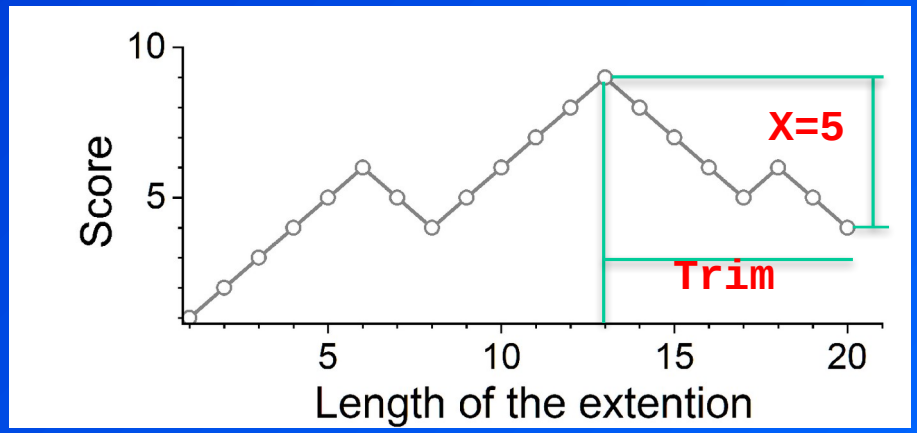
Words triggering extension

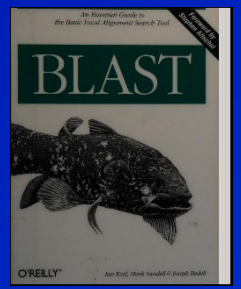
Raw score

Drop-off score

```
The quick brown fox jump
The quiet brown cat purr
123 45654 56789 870 5654
000 00012 10000 123 4345
```

Trim to last maximum





I. Korf, M. Yandell, and J. Bedell. BLAST. O'Reilly Media (2003), pp. 79 - 87

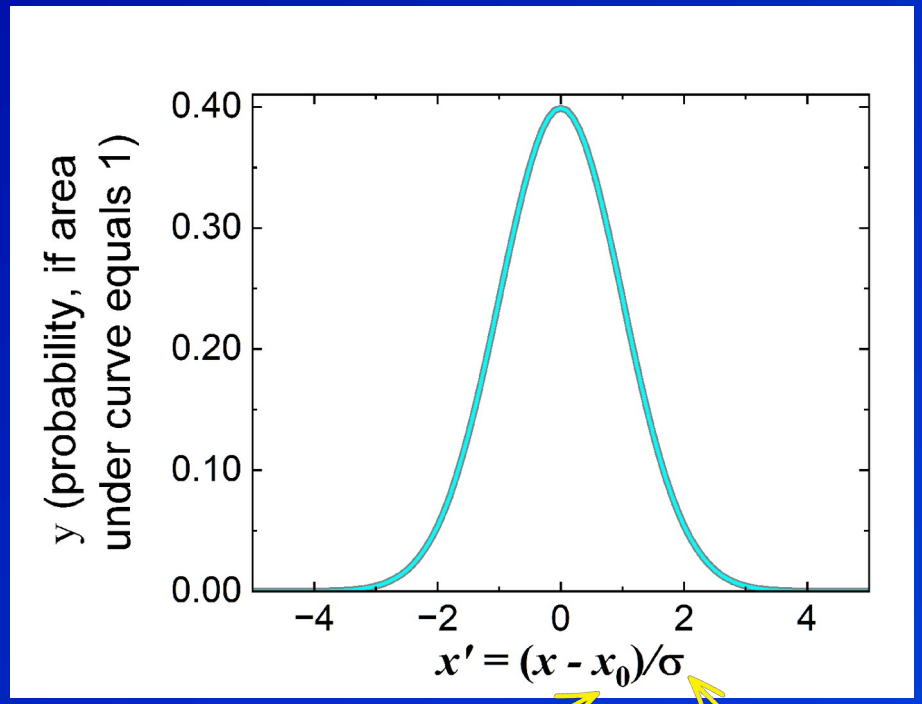
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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Normal distribution

Function



Mean, median, mode

Standard deviation

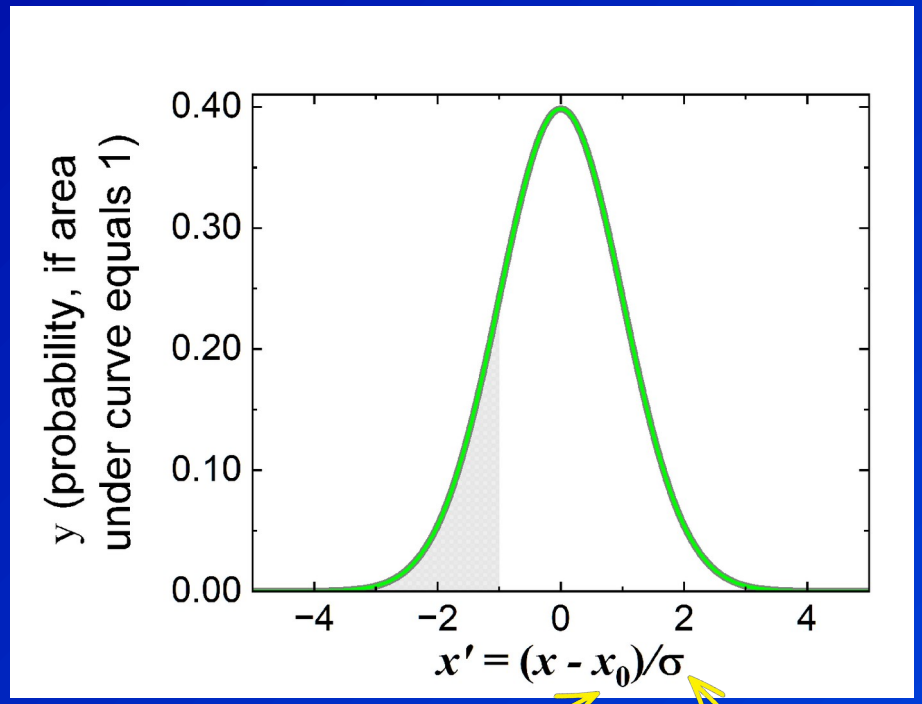
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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Normal distribution

Function



Cumulative distribution function

Mean, median, mode

Standard deviation

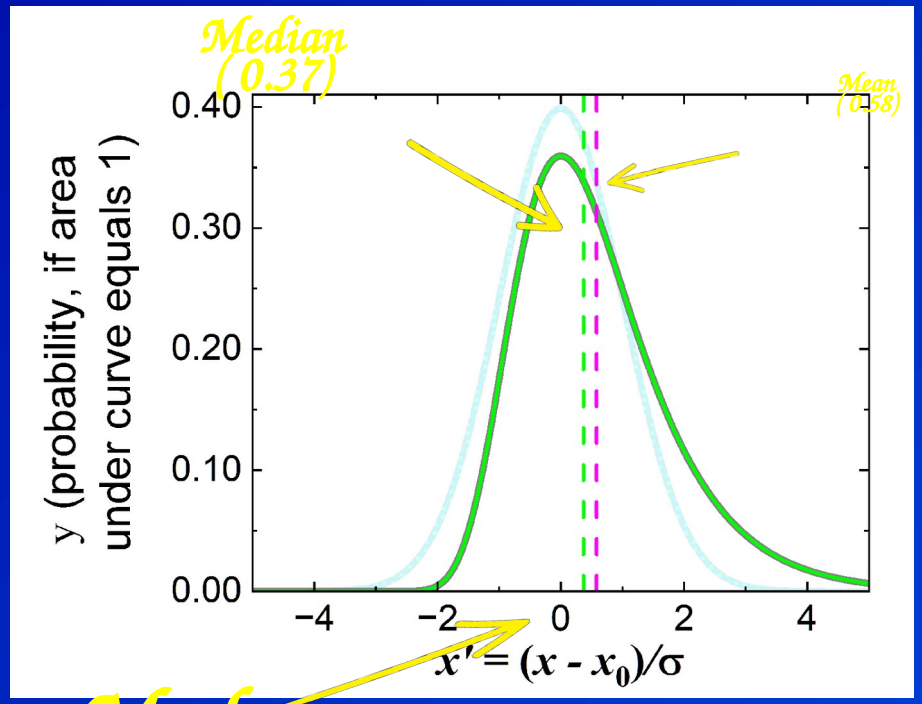
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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Extreme value (Gumbel) distribution

Function



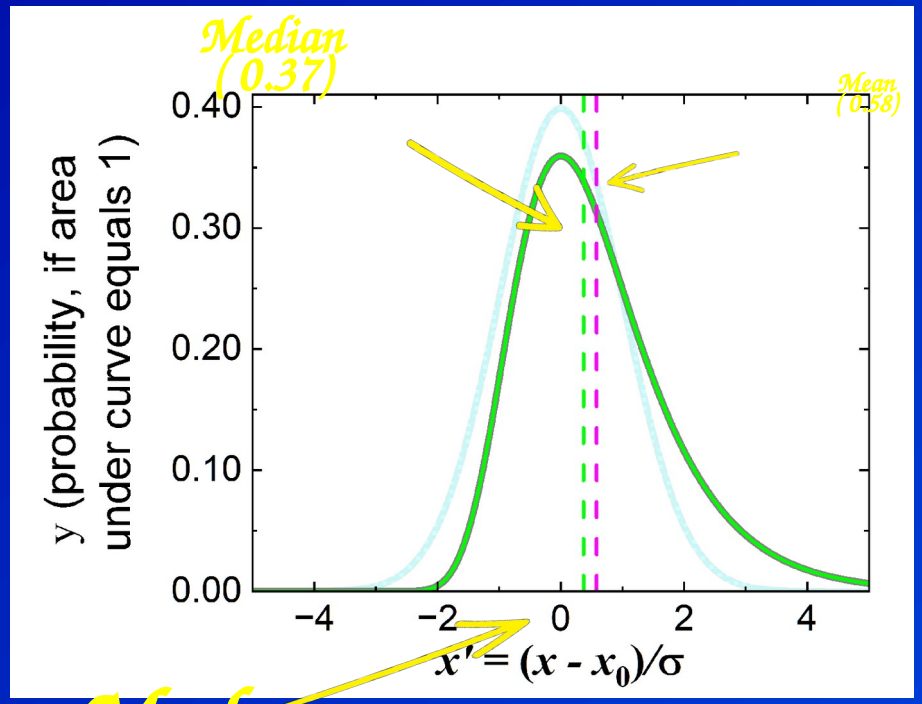
S. Karlin and S.F. Altschul (1990), PNAS, v. 87, pp. 2264 - 2268

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Extreme value (Gumbel) distribution



Function

Cumulative distribution function

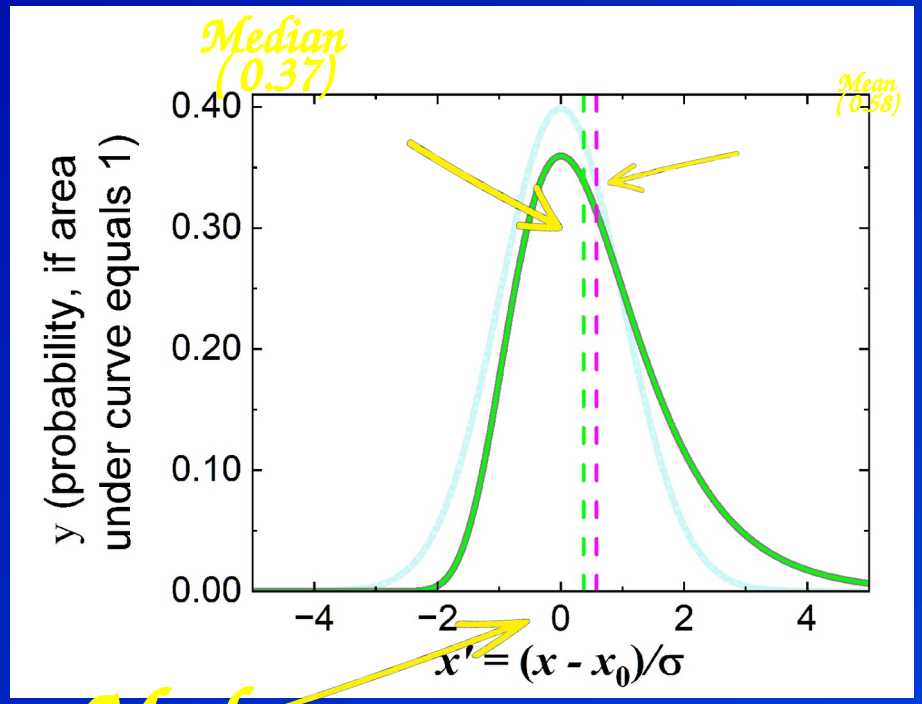
S. Karlin and S.F. Altschul (1990), PNAS, v. 87, pp. 2264 - 2268

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Extreme value (Gumbel) distribution



Function

Cumulative distribution function

S. Karlin and S.F. Altschul (1990), PNAS, v. 87, pp. 2264 - 2268

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Maximum of the extreme value distribution for the BLAST alignments

Total number of letters in the target database

Number of letters in the query

Probability to observe score $x \geq S$ by chance

Expectation value

S. Karlin and S.F. Altschul (1990), PNAS, v. 87, pp. 2264 - 2268

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Expectation value (Karlin-Altschul equation)

Scaling factor for the search space

Scaling factor for the scoring system

$$E = KMNe^{-\lambda S}$$

Search space

Number of high-scoring segment pairs with score S or better expected by chance

K **and** λ **are reported in BLAST output as Karlin-Altschul statistics**

S. Karlin and S.F. Altschul (1990), PNAS, v. 87, pp. 2264 - 2268

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Bit scores

Normalized for a given scoring system and size of the search space

Can be compared between different alignment runs

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BLAST Algorithm

Stage III. Evaluating statistical alignment significance

Effective search space

Average length of the alignments

S. F. Altschul and W. Gish (1996), Methods in Enzymology, v. 266, pp. 460–480.

Stockholms universitet

PSI-BLAST

Position-Specific Iterated BLAST

Differences with ordinary BLAST

Position-specific scoring matrix

Iterative search with PSSM update

Altschul, S. F., Madden, T.L., Schäffer, A.A. et al. (1997), Nuclear Acids Research, v. 25, pp. 3389–3402.

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Position-Specific Scoring Matrix

Algorithm

Construct multiple sequence alignment, MSA

Count residue and gap occurrences in MSA, $N_{i,j}$

Add pseudocounts c_j

Calculate residue frequencies

Estimate background frequencies, q_j

Calculate log-odds scores

Arrange $S_{i,j}$ into a matrix

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Position-Specific Scoring Matrix

Example without gaps

No pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P

Residue counts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	4	1	0	0	0	0	0	0	0
2	2	0	3	0	0	0	0	0	0
3	0	0	0	4	0	0	0	0	1
4	0	0	0	0	4	1	0	0	0
5	0	0	0	0	0	0	4	1	0

Residue frequencies

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	0.80	0.20	0	0	0	0	0	0	0
2	0.40	0	0.60	0	0	0	0	0	0
3	0	0	0	0.80	0	0	0	0	0.20
4	0	0	0	0	0.80	0.20	0	0	0
5	0	0	0	0	0	0	0.80	0.20	0

Background frequencies

natural occurrences

Residue type	Frequency
A	0.074
R	0.042
G	0.074
L	0.076
S	0.081
T	0.062
P	0.050
Q	0.037

Log-odds scores (PSSM)

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	3.43	2.25	-∞	-∞	-∞	-∞	-∞	-∞	-∞
2	2.43	-∞	3.02	-∞	-∞	-∞	-∞	-∞	-∞
3	-∞	-∞	-∞	3.30	-∞	-∞	-∞	-∞	2.40
4	-∞	-∞	-∞	-∞	3.30	1.69	-∞	-∞	-∞
5	-∞	-∞	-∞	-∞	-∞	-∞	4.00	5.41	-∞

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Position-Specific Scoring Matrix

Example without gaps

With uniform pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P

Residue counts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	5	2	1	1	1	1	1	1	1
2	3	1	4	1	1	1	1	1	1
3	1	1	1	5	1	1	1	1	2
4	1	1	1	1	5	2	1	1	1
5	1	1	1	1	1	1	5	2	1

Residue frequencies

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	0.36	0.14	0.07	0.07	0.07	0.07	0.07	0.07	0.07
2	0.21	0.07	0.29	0.07	0.07	0.07	0.07	0.07	0.07
3	0.07	0.07	0.07	0.36	0.07	0.07	0.07	0.07	0.14
4	0.07	0.07	0.07	0.07	0.36	0.14	0.07	0.07	0.07
5	0.07	0.07	0.07	0.07	0.07	0.07	0.36	0.14	0.07

Background frequencies

natural occurrences

Residue type	Frequency
A	0.074
R	0.042
G	0.074
L	0.076
S	0.081
T	0.062
P	0.050
Q	0.037

Log-odds scores (PSSM)

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	2.15	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.85	2.02	0.88

Position-Specific Scoring Matrix

Stockholms universitet

Role of pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P

Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	2.15	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.85	2.02	0.88

Residue not in the alignment

Naturally rare residue

no pseudocounts

Position	Residue type A	R	G	L	S	T	P	Q	I
1	2.43	2.25	-∞	-∞	-∞	-∞	-∞	-∞	-∞
2	2.43	-∞	3.02	-∞	-∞	-∞	-∞	-∞	-∞
3	-∞	-∞	-∞	3.30	-∞	-∞	-∞	-∞	2.40
4	-∞	-∞	-∞	-∞	3.30	1.69	-∞	-∞	-∞
5	-∞	-∞	-∞	-∞	-∞	-∞	4.00	5.41	-∞

Residue type	Frequency
A	0.074
R	0.042
G	0.074
L	0.076
S	0.081
T	0.062
P	0.050
Q	0.037
I	0.038

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Position-Specific Scoring Matrix

Role of pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P



Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	2.15	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.85	2.02	0.88

no pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	3.43	2.25	-∞	-∞	-∞	-∞	-∞	-∞	-∞
2	2.43	-∞	3.02	-∞	-∞	-∞	-∞	-∞	-∞
3	-∞	-∞	-∞	3.30	-∞	-∞	-∞	-∞	2.40
4	-∞	-∞	-∞	-∞	3.30	1.69	-∞	-∞	-∞
5	-∞	-∞	-∞	-∞	-∞	-∞	4.00	5.41	-∞

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Position-Specific Scoring Matrix

Role of pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P
Consensus	A	G	L	S	P

Sequence with maximum score

with uniform pseudocounts

A G L S P

2.28 + 1.97+ 2.24+ 2.15+ 2.85 = 11.49

no pseudocounts

A G L S Q

3.43 + 3.02+ 3.30+ 3.30+ 5.41 = 18.46

Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	2.15	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.85	2.02	0.88

no pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	3.43	2.25	-∞	-∞	-∞	-∞	-∞	-∞	-∞
2	2.43	-∞	3.02	-∞	-∞	-∞	-∞	-∞	-∞
3	-∞	-∞	-∞	3.30	-∞	-∞	-∞	-∞	2.40
4	-∞	-∞	-∞	-∞	3.30	1.69	-∞	-∞	-∞
5	-∞	-∞	-∞	-∞	-∞	-∞	4.00	5.41	-∞

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Position-Specific Scoring Matrix

Example without gaps

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P
Consensus	A	G	L	S	P

Sequence with maximum score

with uniform pseudocounts

A G L S P

2.28 + 1.97+ 2.24+ 2.15 + 2.85 = 11.23

Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	2.15	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.85	2.02	0.88

Arbitrary sequence

R A L S P

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Position-Specific Scoring Matrix

Example without gaps

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P
Consensus	A	G	L	S	P

Sequence with maximum score

with uniform pseudocounts

A G L S P

2.28 + 1.97 + 2.24 + 2.15 + 2.85 = 11.49

Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88
2	1.57	0.74	1.97	2.24	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	-0.12	2.15	0.18	2.85	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	-0.21	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	2.02	0.88

Arbitrary sequence

R A L S P

1.83 + 1.57 + 2.24 + 2.15 + 2.85 = 10.64

Related!

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Position-Specific Scoring Matrix

Example without gaps

Multiple sequence alignment

Position	1	2	3	4	5
Protein 1	A	G	L	S	P
Protein 2	A	G	L	T	P
Protein 3	R	G	I	S	P
Protein 4	A	A	L	S	Q
Protein 5	A	A	L	S	P
Consensus	A	G	L	S	P

Sequence with maximum score

with uniform pseudocounts

A G L S P

2.28 + 1.97+ 2.24+ 2.15+ 2.85= 11.49

Log-odds scores (PSSM)

with uniform pseudocounts

Position	Residue type								
	A	R	G	L	S	T	P	Q	I
1	2.28	1.83	1.97	-0.12	-0.21	0.18	0.49	0.92	0.88
2	-0.08	0.74	-0.08	2.24	-0.21	0.18	0.49	0.92	0.88
3	-0.08	0.74	-0.08	-0.12	2.15	-0.08	2.85	0.92	1.98
4	-0.08	0.74	-0.08	-0.12	-0.21	1.27	0.49	0.92	0.88
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	-0.08	2.02	0.88

Arbitrary sequence

R A L S P

1.83 + 1.57+ 2.24+ 2.15+ 2.85= 10.64

Related!

G G T A A

-0.08+ 1.97+ 0.18- 0.08- 0.08= 1.91

Not related!

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Position-Specific Scoring Matrix

Example with gaps

With uniform pseudocounts

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P

Residue counts

Gap

Position	A	R	G	L	S	T	P	Q	I	-
1	5	2	1	1	1	1	1	1	1	1
2	3	1	4	1	1	1	1	1	1	1
3	1	1	1	4	1	1	1	1	2	2
4	1	1	1	1	5	2	1	1	1	1
5	1	3	1	1	1	1	1	1	1	4

Residue frequencies

Position	A	R	G	L	S	T	P	Q	I	-
1	0.33	0.13	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
2	0.20	0.07	0.27	0.07	0.07	0.07	0.07	0.07	0.07	0.07
3	0.07	0.07	0.07	0.27	0.07	0.07	0.07	0.07	0.13	0.13
4	0.07	0.07	0.07	0.07	0.33	0.13	0.07	0.07	0.07	0.07
5	0.07	0.20	0.07	0.07	0.07	0.07	0.07	0.13	0.07	0.27
6	0.07	0.07	0.07	0.07	0.07	0.07	0.33	0.13	0.07	0.07

Background frequencies

natural occurrences

Residue type	Frequency
A	0.074
R	0.042
G	0.074
L	0.076
S	0.081
T	0.062
P	0.050
Q	0.037
I	0.038

Log-odds scores (PSSM)

Frequency in MSA

Position	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.77	-0.03
4	-0.08	0.74	-0.08	-0.12	2.03	0.68	0.49	0.92	0.88	-0.93
5	-0.08	2.25	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P



Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.77	-0.03
4	-0.08	0.74	-0.08	-0.12	2.03	0.68	0.49	0.92	0.88	-0.93
5	-0.08	2.25	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	0.74	-0.08	-0.12	2.03	0.18	0.49	0.92	0.77	-0.03
4	-0.08	0.74	-0.08	-0.12	-0.21	0.68	0.49	0.92	0.88	-0.93
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	-0.93

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Log-odds scores (PSSM)

Position	Residue type								
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	2.25	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.77	-0.03
4	-0.08	0.74	-0.08	-0.12	-0.21	0.68	0.49	0.92	0.88	-0.93
5	-0.08	-0.08	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	-0.93

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	1.87	-0.08	1.83	-0.21	0.18	0.49	0.92	0.77	-0.03
4	1.43	0.74	-0.08	-0.12	2.03	0.18	0.49	0.92	0.88	-0.93
5	-0.08	-0.08	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	-0.93

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Arbitrary sequence of length 5

R A S R P

-RASRP -0.93+0.74-0.08+2.03+2.25+2.72=6.72

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Log-odds scores (PSSM)

Position	Residue type A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	-0.08	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.88	-0.93
3	1.43	-0.08	-0.08	-0.12	2.03	0.18	0.49	0.92	0.88	
4	-0.08	2.25	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.77	-0.03
5	-0.08	0.74	-0.08	-0.12	-0.21	0.68	2.72	0.92	0.88	-0.93
6	-0.08	-0.08	-0.08	-0.12	-0.21	0.18	0.49	1.81	0.88	-0.93

Arbitrary sequence of length 5

R A S R P

-RASRP -0.93+0.74-0.08+2.03+2.25+2.72=6.72

R-ASRP 1.63-0.93-0.08+2.03+2.25+2.72=6.72

Position-Specific Scoring Matrix

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Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	1.87	-0.08	-0.12	2.03	0.18	0.49	0.92	0.88	-0.03
4	-0.08	1.87	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.77	
5		0.74	-0.08	-0.12	-0.21	0.68	0.49	0.92	0.88	-0.93
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Arbitrary sequence of length 5

R A S R P

-RASRP -0.93+0.74-0.08+2.03+2.25+2.72=6.72

R-ASRP 1.63-0.93-0.08+2.03+2.25+2.72=6.72

RA-SRP 1.63+1.43-0.03+2.03+2.25+2.72=10.03

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	-1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	0.74	-0.08	-0.12	2.03	0.18	0.49	0.92	0.77	-0.03
4	-0.08	0.74	-0.08	-0.12	-0.21	0.68	0.49	0.92	0.88	-0.93
5	-0.08	-0.08	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-1.02
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18		1.81	0.88	-0.93

Arbitrary sequence of length 5

R A S R P

-RASRP -0.93+0.74-0.08+2.03+2.25+2.72=6.72

R-ASRP 1.63-0.93-0.08+2.03+2.25+2.72=6.72

RA-SRP 1.63+1.43-0.03+2.03+2.25+2.72=10.03

RAS-RP 1.63+1.43-0.21-0.93+2.25+2.72=6.89

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	2.16	1.63	1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	1.43	0.74	-0.08	1.83	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	0.74	-0.08	-0.12	2.03	0.18	0.49	0.92	0.88	-0.03
4	-0.08	2.25	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.77	1.02
5			-0.08	-0.12		0.68	0.49	0.92	0.88	
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93

Arbitrary sequence of length 5

R A S R P

-RASRP	-0.93+0.74-0.08+2.03+2.25+2.72=6.72
R-ASRP	1.63-0.93-0.08+2.03+2.25+2.72=6.72
RA-SRP	1.63+1.43-0.03+2.03+2.25+2.72=10.03
RAS-RP	1.63+1.43-0.21-0.93+2.25+2.72=6.89
RASR-P	**1.63+1.43-0.21+0.74+1.02+2.72=7.33**

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Log-odds scores (PSSM)

Position	Residue type									
	A	R	G	L	S	T	P	Q	I	-
1	1.43	0.74	-1.87	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	-0.08			-0.12	-0.21	0.18	0.49	0.92	0.88	-0.03
3		2.25	-0.08			0.18	0.49	0.92	0.77	1.02
4			-0.08	-0.12		0.68	0.49	0.92	0.88	-0.93
5	-0.08		-0.08	-0.12	-0.21	0.18		1.81	0.88	
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18		1.81	0.88	

Arbitrary sequence of length 5

R A S R P

-RASRP	-0.93+0.74-0.08+2.03+2.25+2.72=6.72
R-ASRP	1.63-0.93-0.08+2.03+2.25+2.72=6.72
RA-SRP	1.63+1.43-0.03+2.03+2.25+2.72=10.03
RAS-RP	1.63+1.43-0.21-0.93+2.25+2.72=6.89
RASR-P	1.63+1.43-0.21+0.74+1.02+2.72=7.33
RASRP-	1.63+1.43-0.21+0.74+0.49-0.93=3.15

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Position-Specific Scoring Matrix

Example with gaps

Multiple sequence alignment

Position	1	2	3	4	5	6
Protein 1	A	G	L	S	-	P
Protein 2	A	G	L	T	R	P
Protein 3	R	G	I	S	R	P
Protein 4	A	A	L	S	-	Q
Protein 5	A	A	-	S	-	P
Consensus	A	G	L	S	R	P

Log-odds scores (PSSM)

Position	Residue type A	R	G	L	S	T	P	Q	I	-
1	1.43	0.74	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
2	-0.08	0.74	-0.08	-0.12	-0.21	0.18	0.49	0.92	0.88	-0.93
3	-0.08	2.25	-0.08	-0.12	2.03	0.18	0.49	0.92	0.77	1.02
4	-0.08	0.74	-0.08	-0.12	-0.21	0.68	0.49	0.92	0.88	-0.93
5	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93
6	-0.08	0.74	-0.08	-0.12	-0.21	0.18	2.72	1.81	0.88	-0.93

Sequence with maximum score

A G L S R P

2.16 + 1.87 + 1.83 + 2.03 + 2.25 + 2.72 = 12.86

Arbitrary sequence of length 6

R G A A P P

1.63 + 1.87 - 0.08 - 0.08 + 0.49 + 2.72 = 6.55

Arbitrary sequence of length 5

R A S R P

-RASRP -0.93+0.74-0.08+2.03+2.25+2.72=6.72 *Max*

R-ASRP 1.63-0.93-0.08+2.03+2.25+2.72=6.72

RA-SRP 1.63+1.43-0.03+2.03+2.25+2.72=10.03

RAS-RP 1.63+1.43-0.21-0.93+2.25+2.72=6.89

RASR-P 1.63+1.43-0.21+0.74+1.02+2.72=7.33

RASRP- 1.63+1.43-0.21+0.74+0.49-0.93=3.15

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Profile Hidden Markov Model

Key concepts

Hidden states

Represents underlying mechanism that generates observable data

***Match,* M** ***Insert,* I** ***Delete,* D**

Transitions

Corresponds to biological events in sequences

***Match-to-Match,* MM** ***Match-to-Insert,* MI** ***Match-to-Delete,* MD**

***Insert-to-Match,* IM** ***Insert-to-Insert,* II** ***Insert-to-Delete,* ID**

***Delete-to-Match,* DM** ***Delete-to-Insert,* DI** ***Delete-to-Delete,* DD**

Emissions

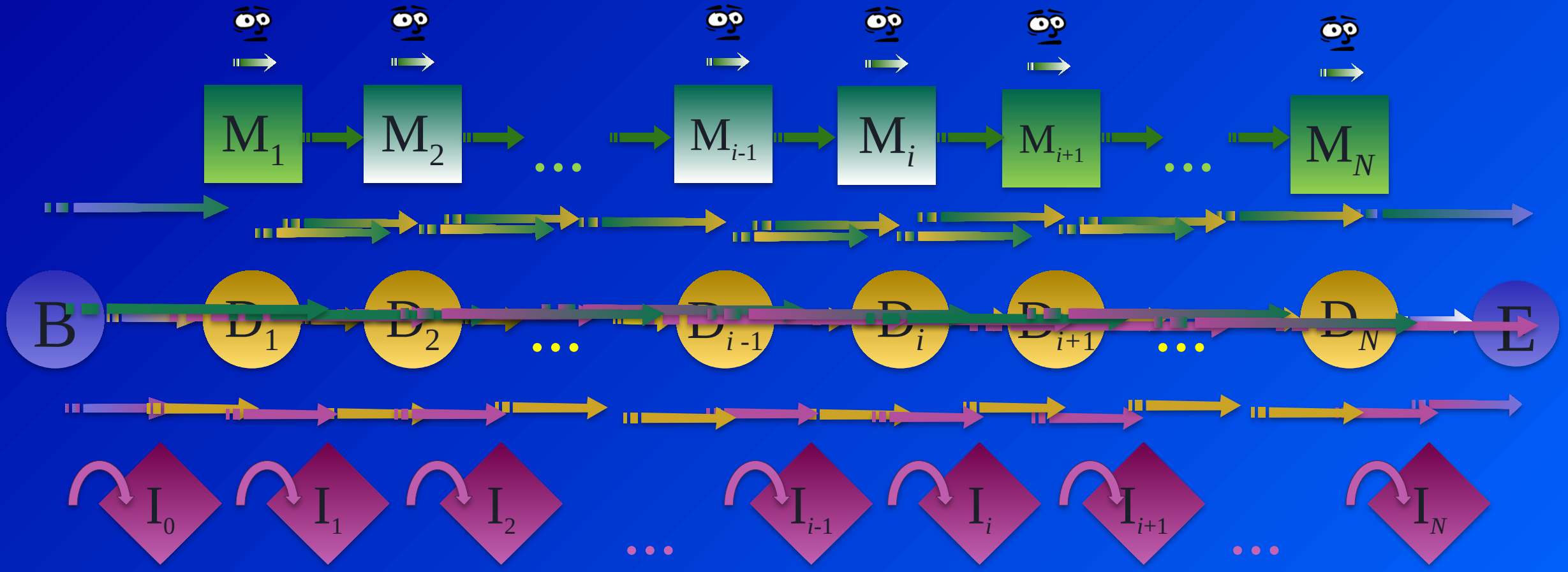
Represents actual observable data in a particular sequence

***Residues/nucleotides in* M, EM(X)** ***Residues/nucleotides in* I, EI(X)**

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Profile Hidden Markov Model

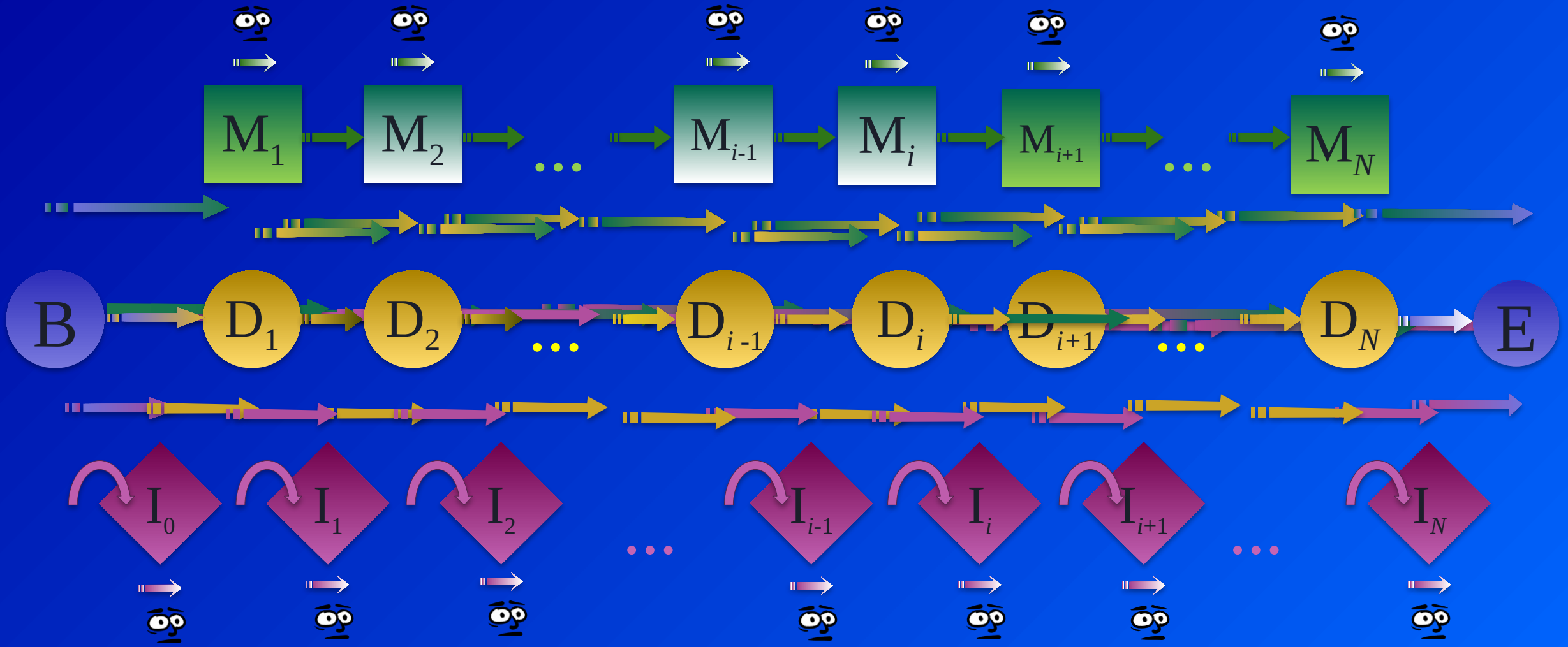
General architecture



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Profile Hidden Markov Model

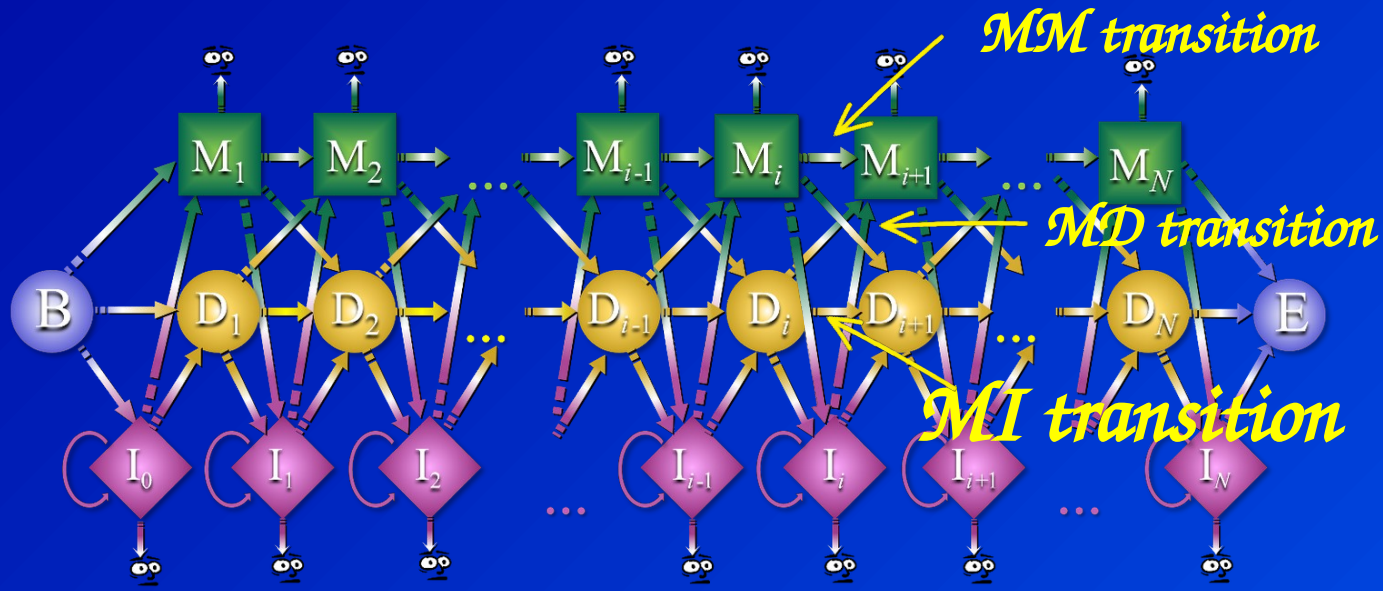
General architecture



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Profile Hidden Markov Model

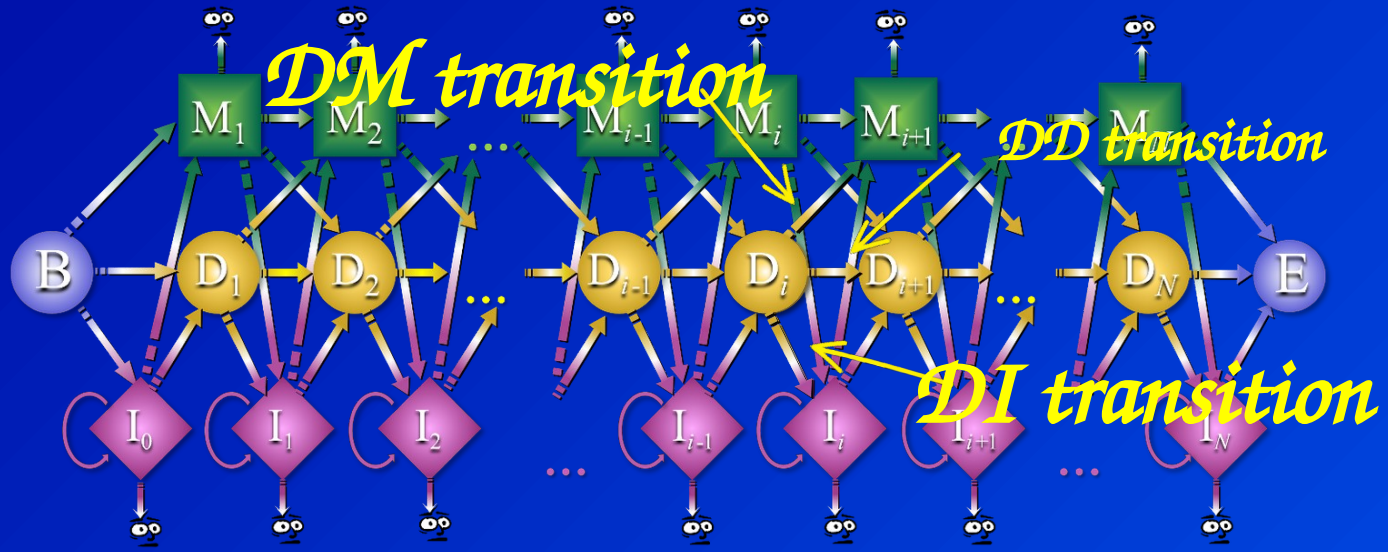
Probabilistic nature



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Profile Hidden Markov Model

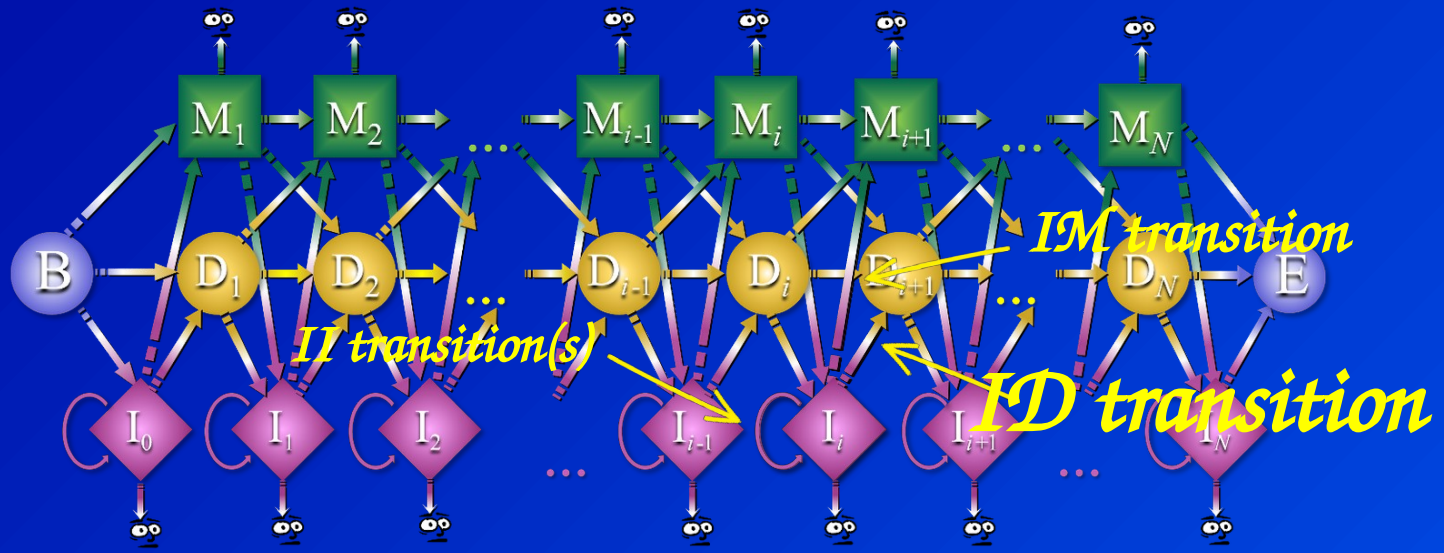
Probabilistic nature



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Profile Hidden Markov Model

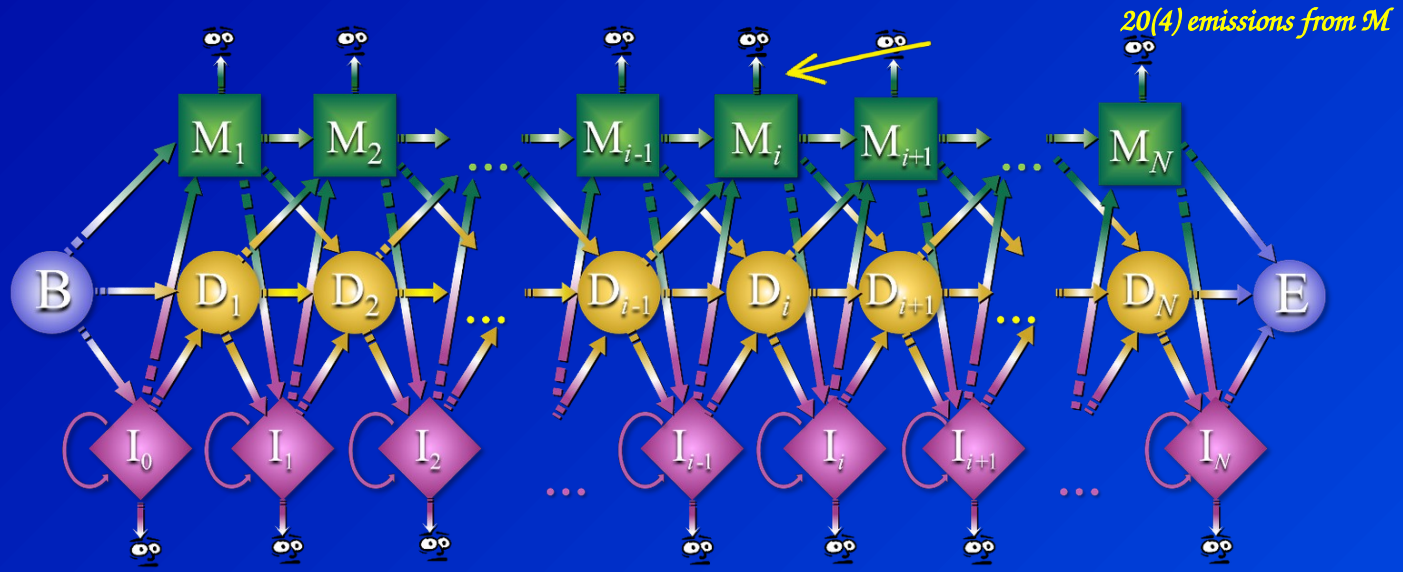
Probabilistic nature



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Profile Hidden Markov Model

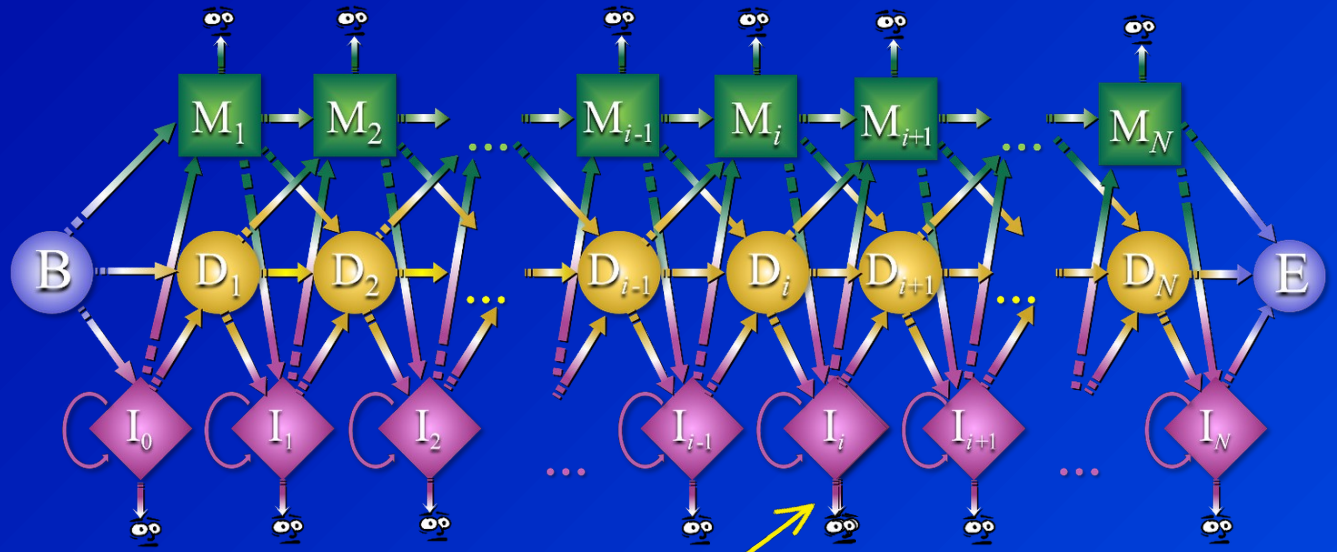
Probabilistic nature



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Profile Hidden Markov Model

Probabilistic nature



20 (4) emissions from I

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	-	-	-	K	-	D
Protein 2	M	A	K	L	T	-	-	-	R	P	-
Protein 3	-	G	K	S	R	-	-	-	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	-	-	-	H	W	Q

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	-	-	-	K	-	D
Protein 2	M	A	K	L	T	-	-	-	R	P	-
Protein 3	-	G	K	S	R	-	-	-	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	Q	N	A	H	W	Q

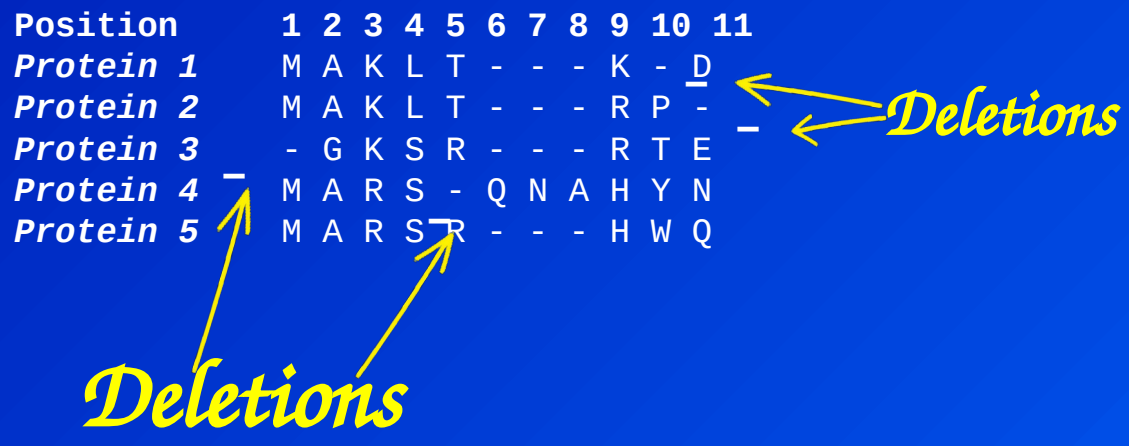
Insertion

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment



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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	-	-	-	K	-	D
Protein 2	M	A	K	L	T	-	-	-	R	P	-
Protein 3	-	G	K	S	R	-	-	-	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	-	-	-	H	W	Q

Not deletions!

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q
						.	.	.			

Not deletions!

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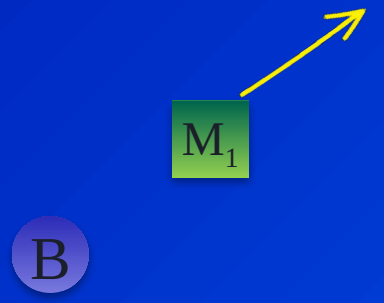
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Define Match states

Position		1	2	3	4	5	6	7	8	9	10	11
	1											
Protein 1	M	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	M	A	K	L	T	.	.	.	R	P	-
Protein 3	M	-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	M	A	R	S	R	.	.	.	H	W	Q
	M											



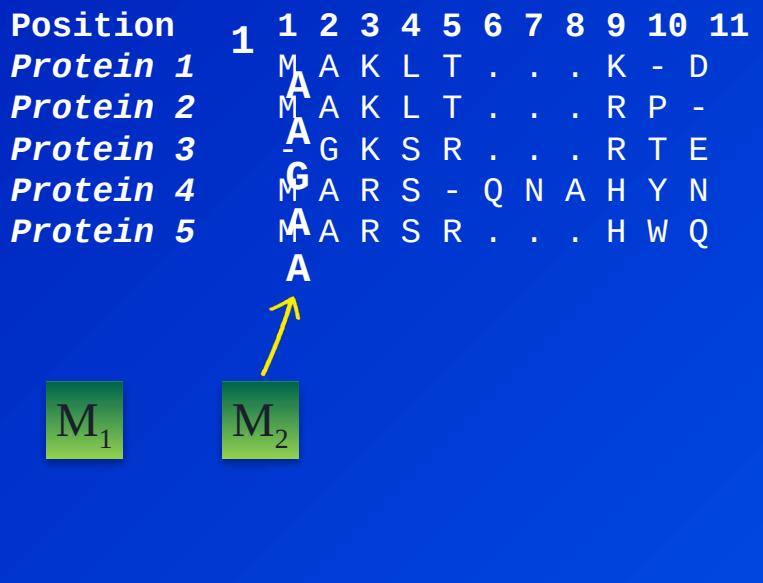
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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Define Match states



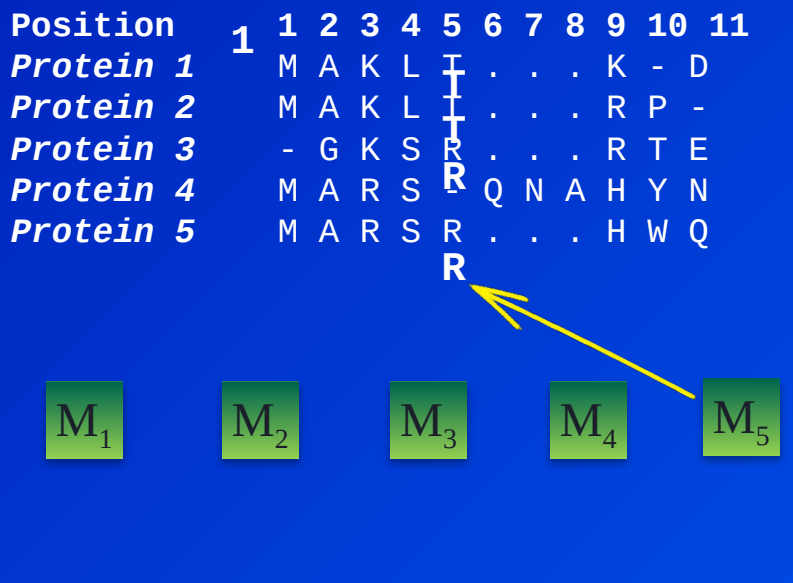
Stockholms universitet

Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Define Match states



Profile Hidden Markov Model

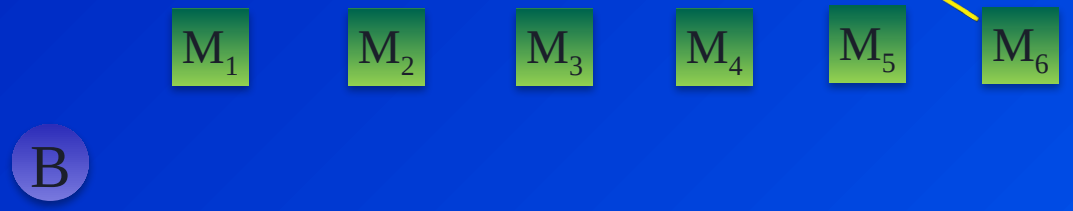
Stockholms universitet

Constructing (training) HMM profile

Multiple sequence alignment

Define Match states

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	R	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q
											H	



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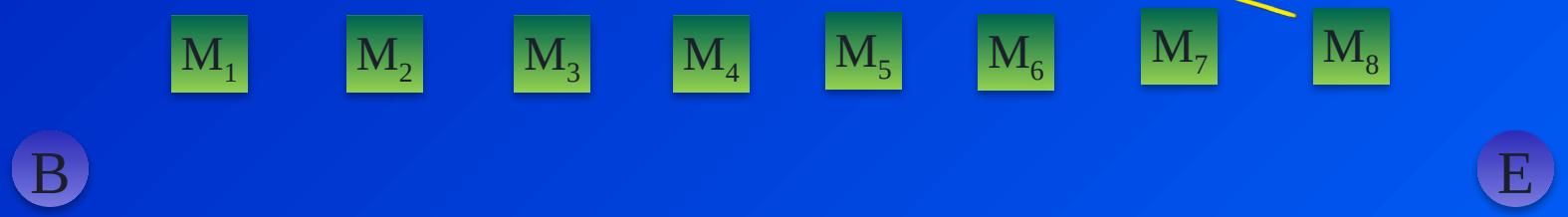
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11	
Protein 1		M	A	K	L	T	.	.	.	K	-	D	
Protein 2		M	A	K	L	T	.	.	.	R	P	-	**D**
Protein 3		-	G	K	S	R	.	.	.	R	T	E	
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N	**E**
Protein 5		M	A	R	S	R	.	.	.	H	W	Q	**N**
													Q

Define Match states



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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Define Match states

Add Delete states



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Profile Hidden Markov Model

Constructing (training) HMM profile

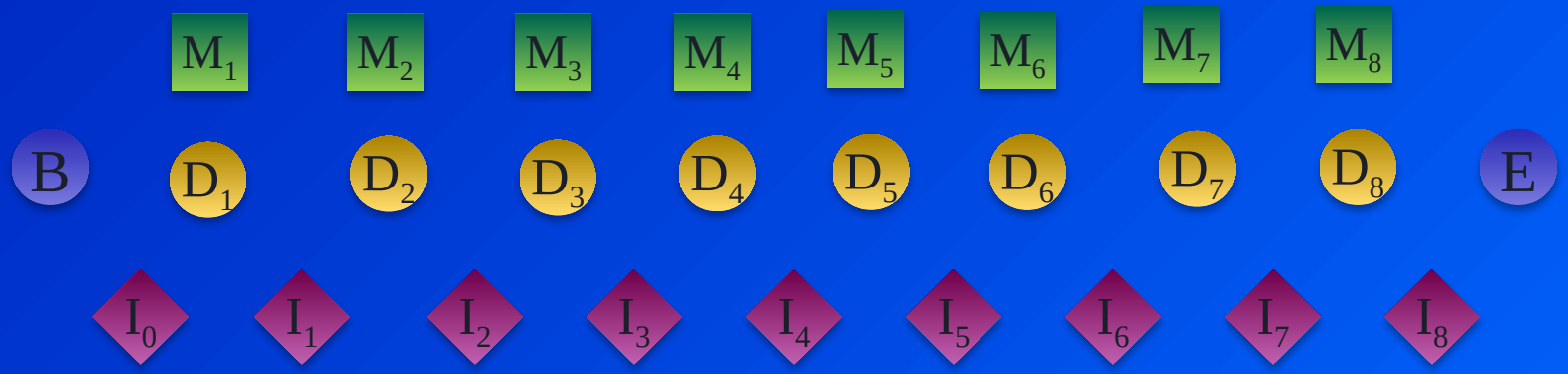
Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Define Match states

Add Delete states

Add Insert states



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Profile Hidden Markov Model

Constructing (training) HMM profile

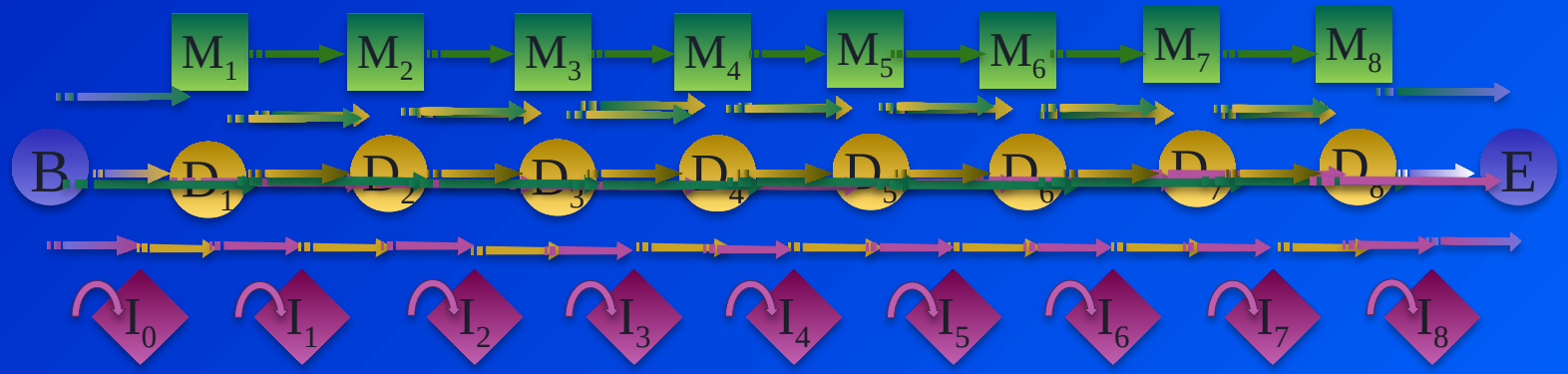
Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Define Match states

Add Delete states

Add Insert states



Add transitions between the states

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	-	-	-	K	-	D
Protein 2		M	A	K	L	T	-	-	-	R	P	-
Protein 3		-	G	K	S	R	-	-	-	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	-	-	-	H	W	Q

Path in the profile for:

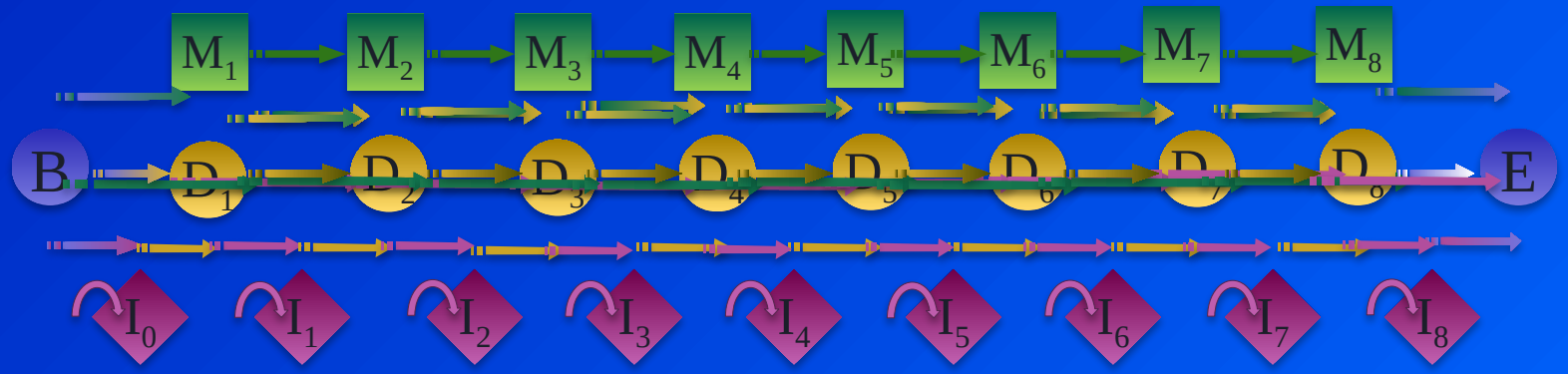
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
	1										
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q
	M	**A**	**K**	**L**	**T**	**.**	**.**	**.**	**K**	**-**	**D**

Path in the profile for:

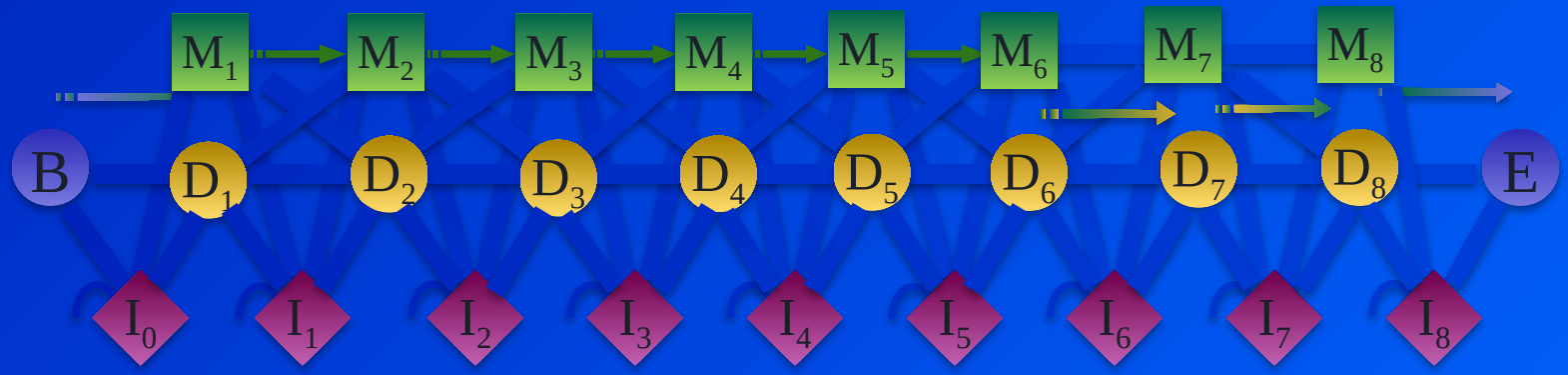
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

M A K L T . . . R P -

Path in the profile for

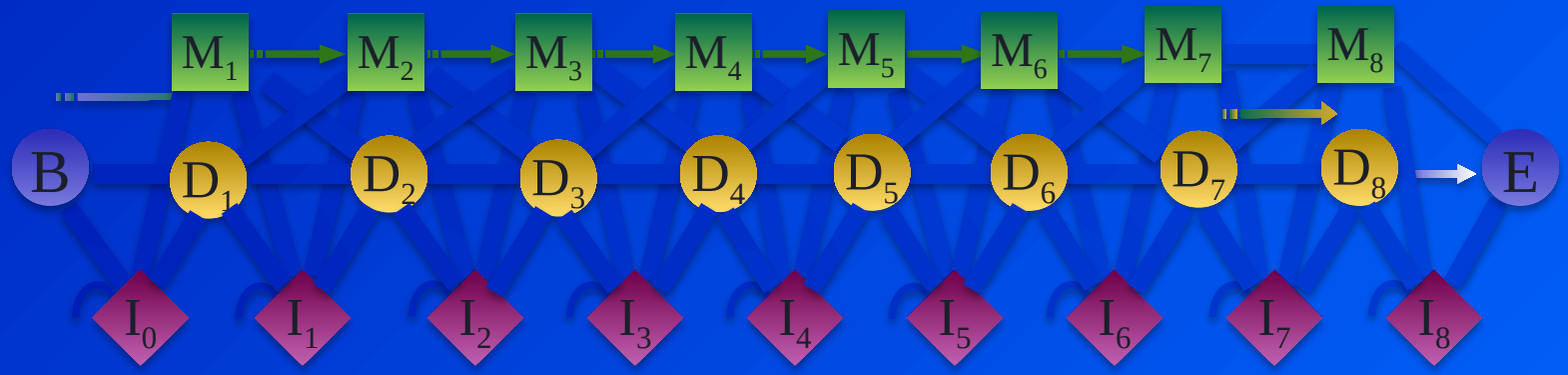
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

- G K S R . . . R T E

Path in the profile for

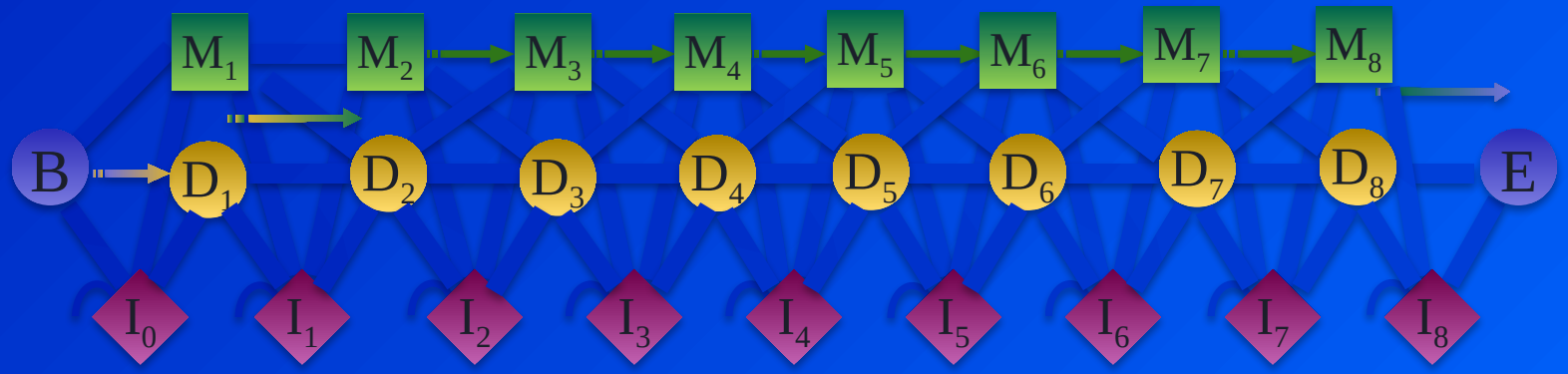
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



Profile Hidden Markov Model

Stockholms universitet

Constructing (training) HMM profile

Multiple sequence alignment

```
Position     1  1 2 3 4 5 6 7 8 9 10 11
Protein 1       M A K L T . . . K - D
Protein 2       M A K L T . . . R P -
Protein 3       - G K S R . . . R T E
Protein 4       M A R S - Q N A H Y N
Protein 5       M A R S R . . . H W Q
M A R S - Q N A H Y N
```

Path in the profile for

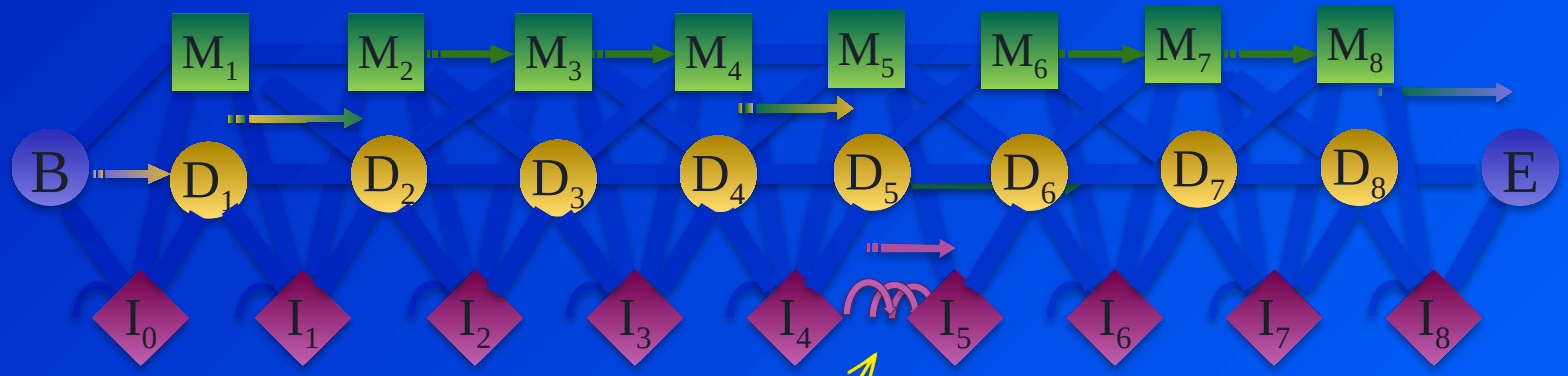
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



3 inserted residues = 3 II transitions

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Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

```
Position      1  1 2 3 4 5 6 7 8 9 10 11
Protein 1        M A K L T . . . K - D
Protein 2        M A K L T . . . R P -
Protein 3        - G K S R . . . R T E
Protein 4        M A R S - Q N A H Y N
Protein 5        M A R S R . . . H W Q
                 M A R S R . . . H W Q
```

Path in the profile for

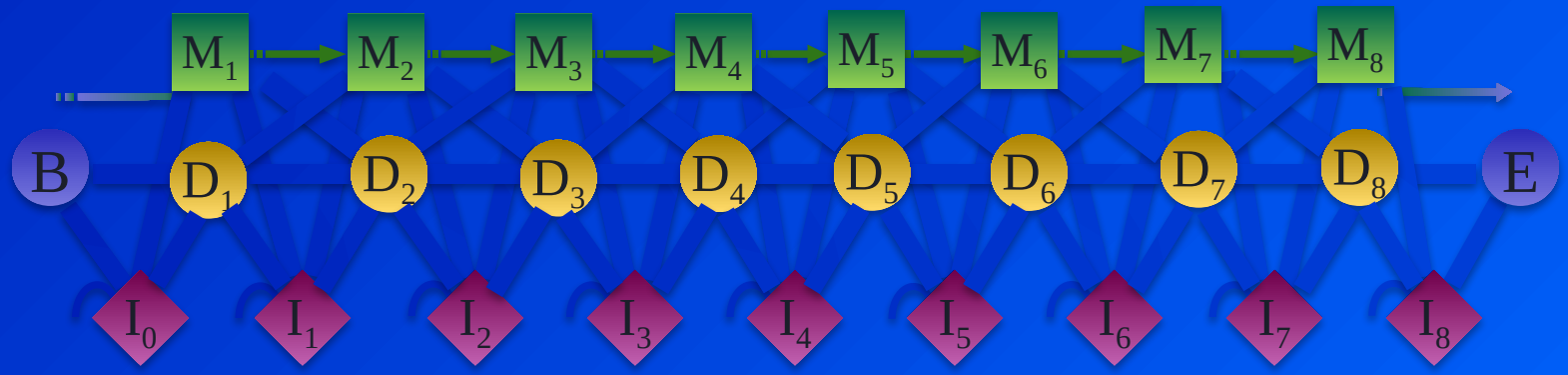
sequence 1

sequence 2

sequence 3

sequence 4

sequence 5



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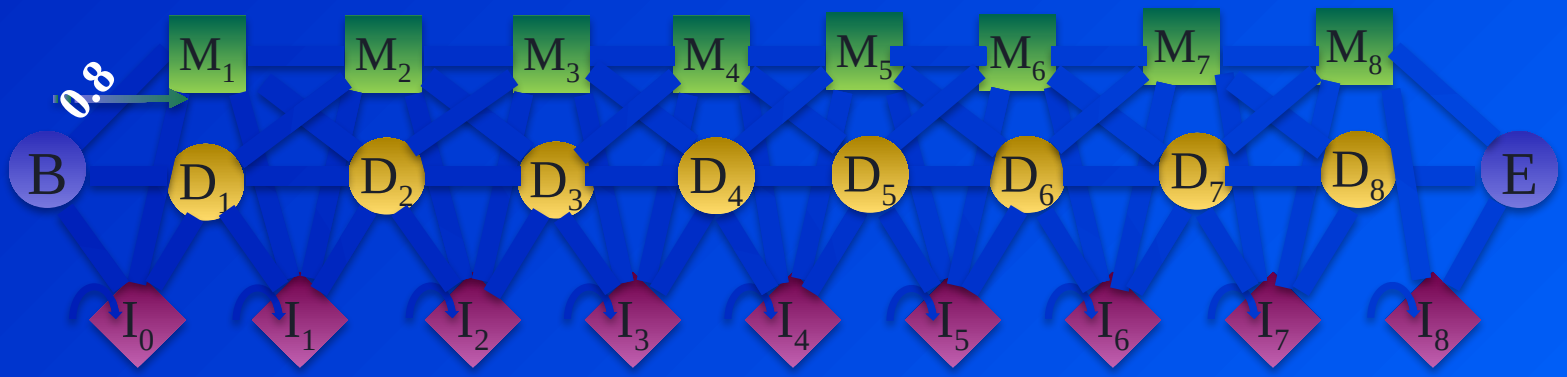
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with a residue in M1

Total number of sequences in MSA

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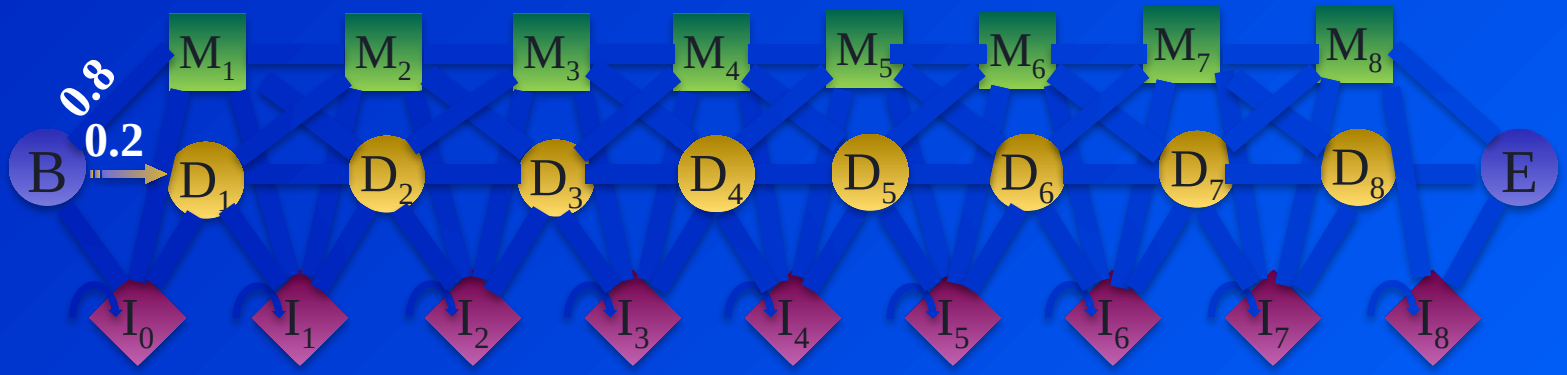
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4	-	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with a gap in M1

Total number of sequences in MSA

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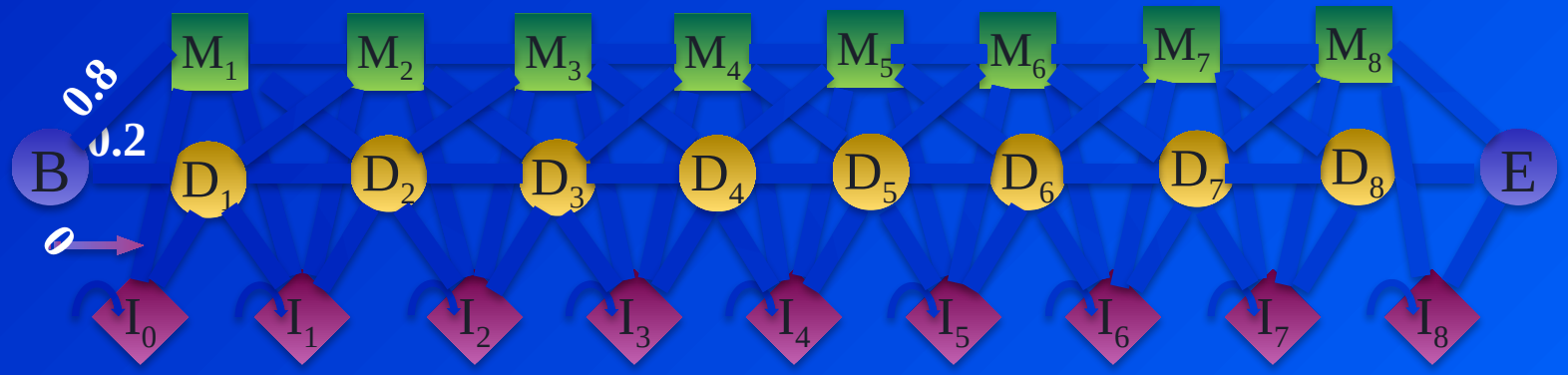
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with an insertion before M1

Total number of sequences in MSA

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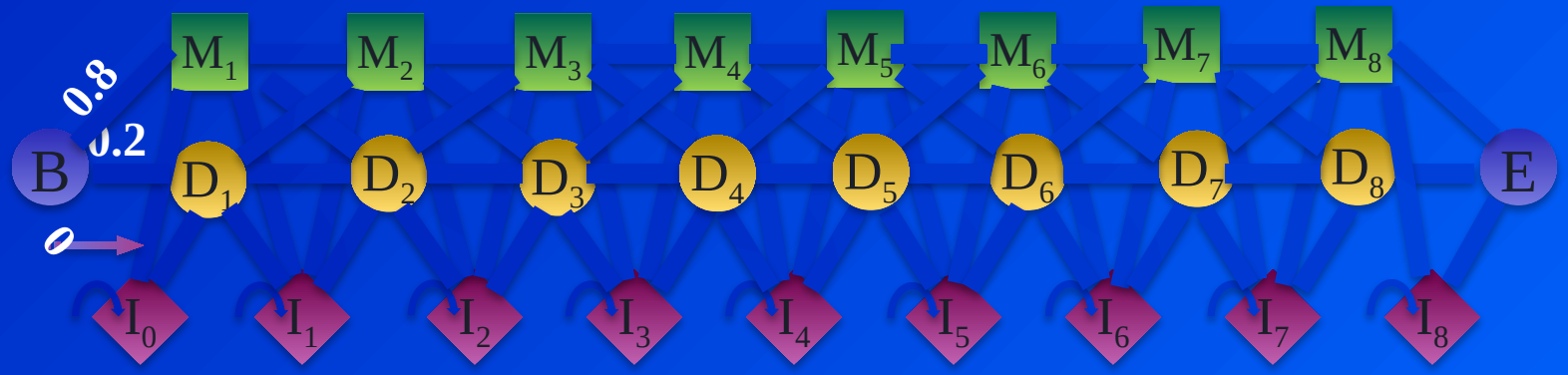
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	1	2	3	4	5	6	7	8	9	10	11
Protein 1		M	A	K	L	T	.	.	.	K	-	D
Protein 2		M	A	K	L	T	.	.	.	R	P	-
Protein 3		-	G	K	S	R	.	.	.	R	T	E
Protein 4		M	A	R	S	-	Q	N	A	H	Y	N
Protein 5		M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with an insertion before M1

Total number of sequences in MSA

Stockholms universitet

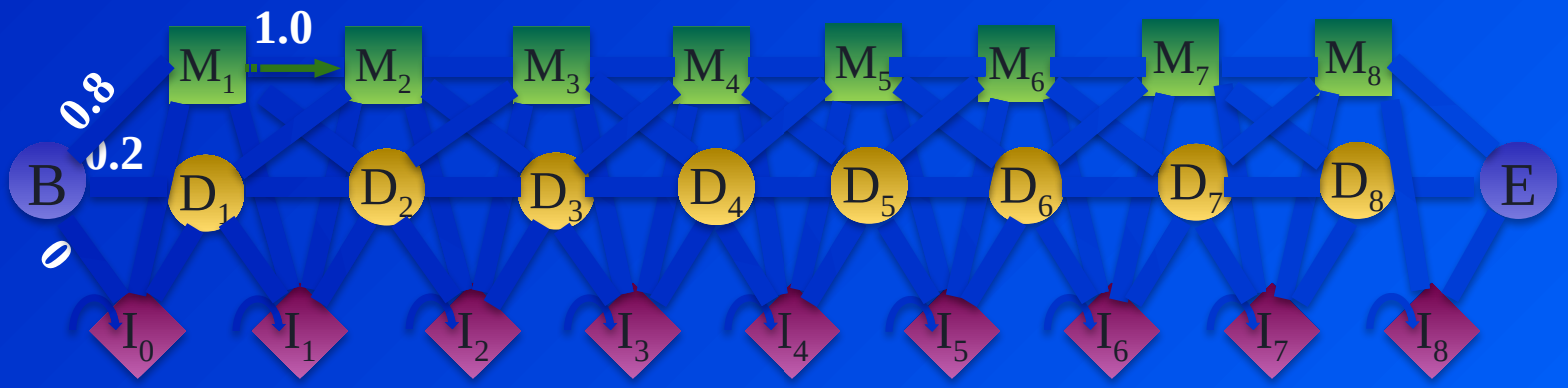
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with a residue in both M1 and M2

Total number of sequences in MSA with a residue in M1

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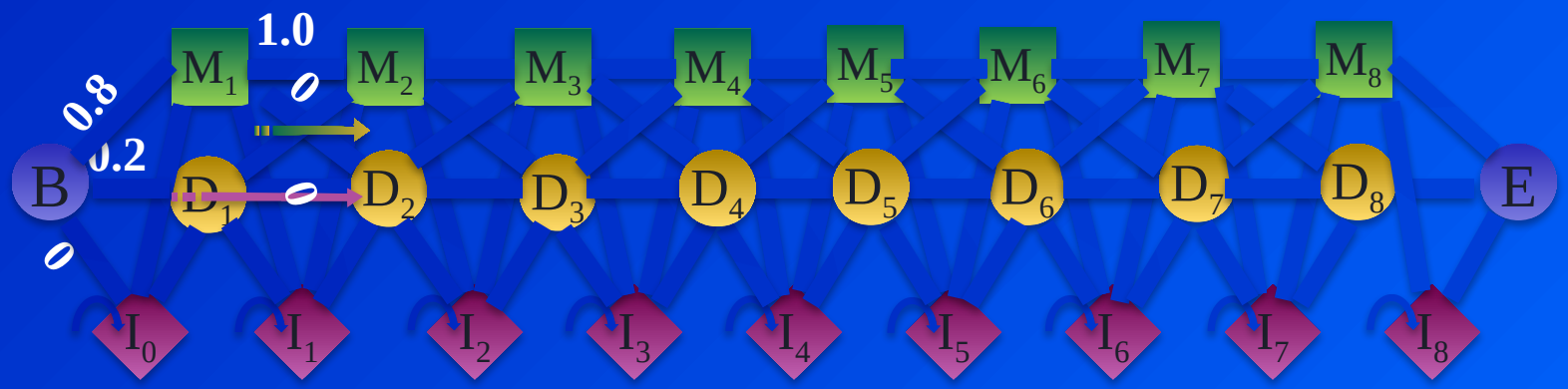
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



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Profile Hidden Markov Model

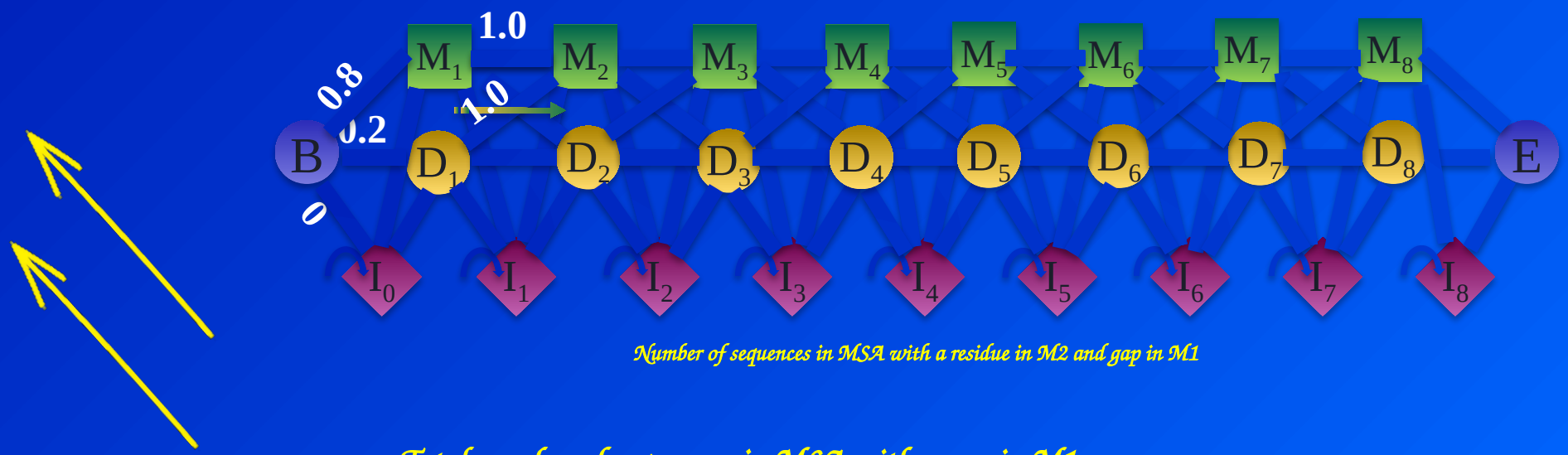
Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Protein 3 - G

Calculate transition probabilities



Number of sequences in MSA with a residue in M2 and gap in M1

Total number of sequences in MSA with a gap in M1

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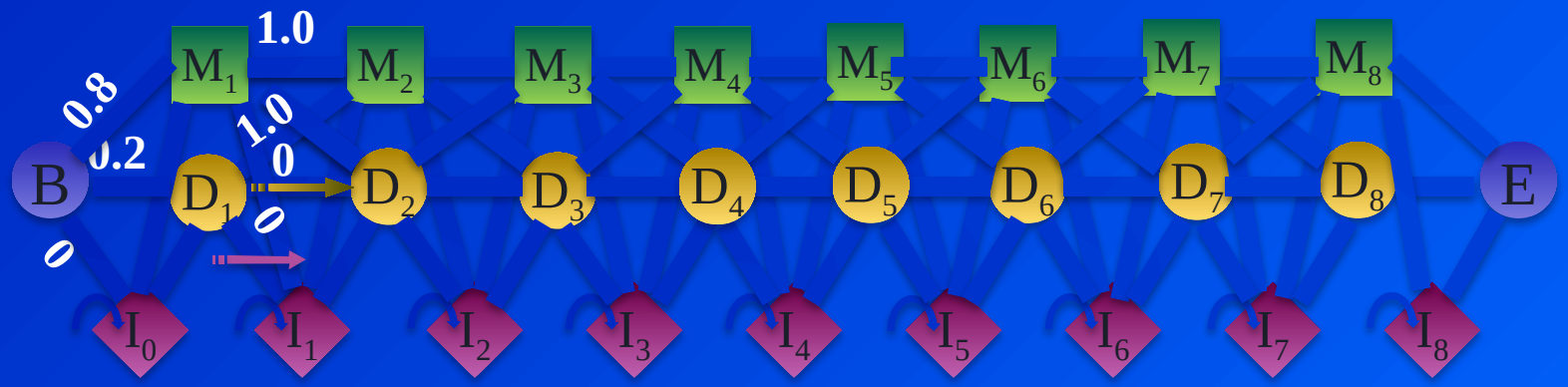
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Calculate transition probabilities



Profile Hidden Markov Model

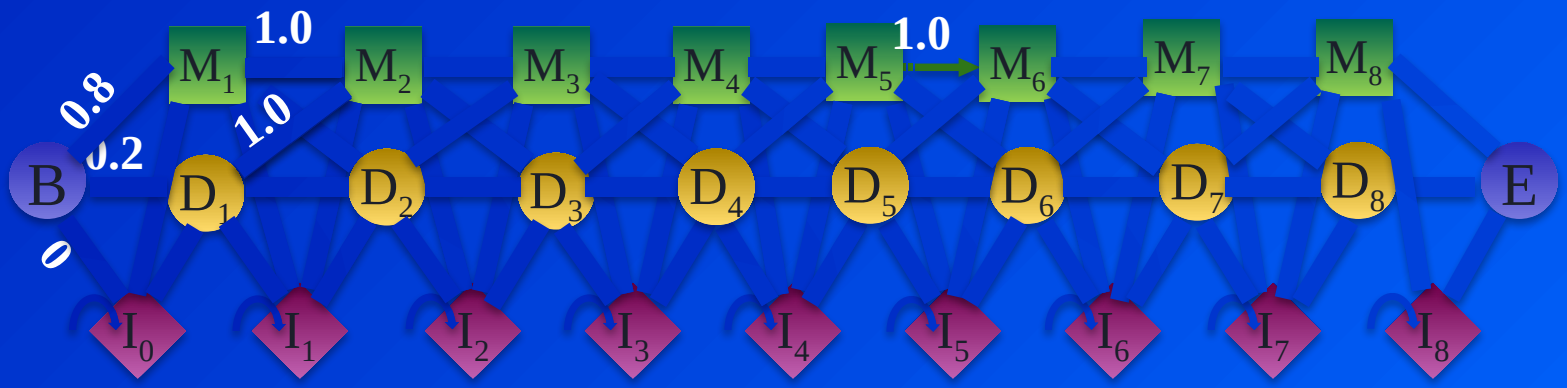
Stockholms universitet

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	I	.	.	.	K	-	D
Protein 2	M	A	K	L	I	.	.	.	R	D	-
Protein 3	-	G	K	S	R	.	.	.	R	I	E
Protein 4	M	A	R	S	R	Q	N	A	H	K	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q
					R				H		

Calculate transition probabilities



Number of sequences in MSA with a residue in both M5 and M6

Total number of sequences in MSA with a residue in M5

Stockholms universitet

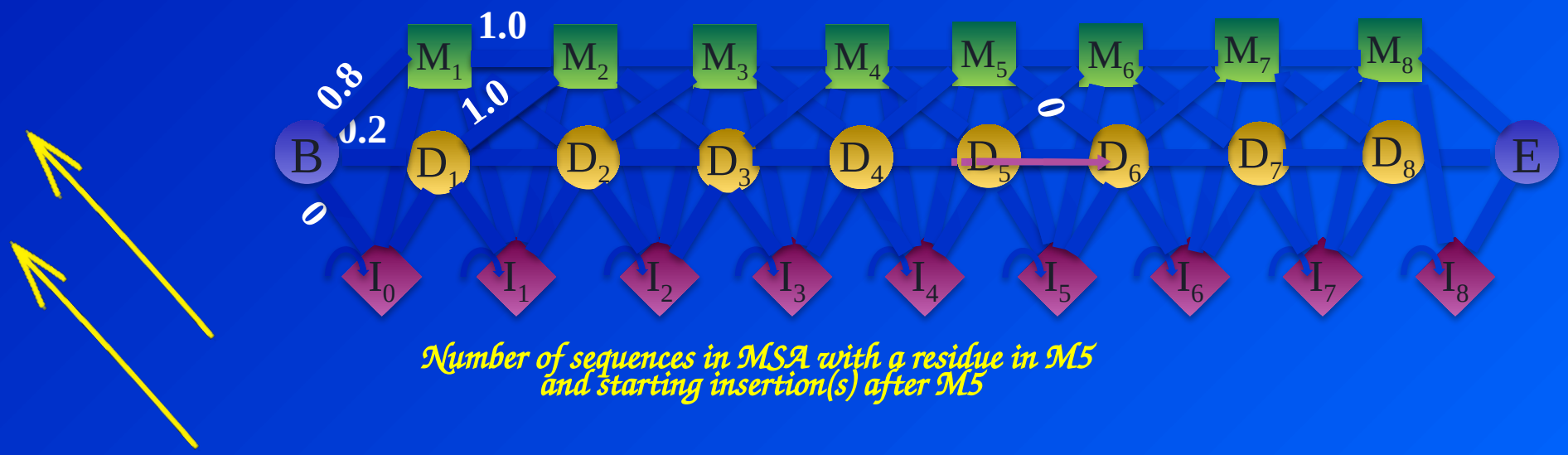
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	Q	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with a residue in M5 and starting insertion(s) after M5

Total number of sequences in MSA with a residue in M5

Stockholms universitet

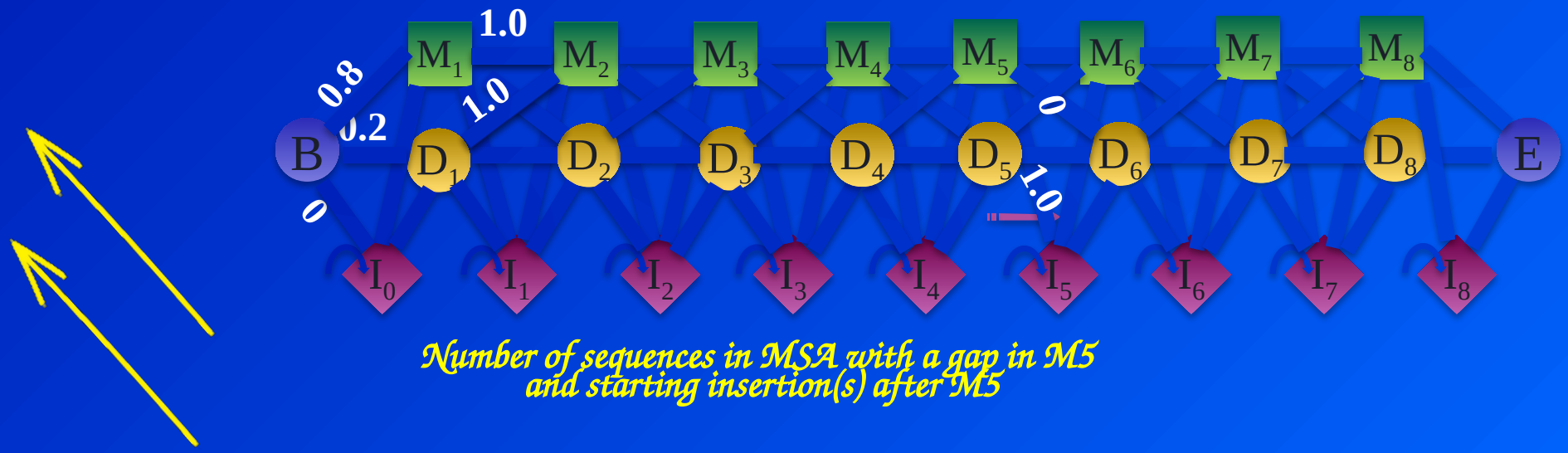
Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	Q	.	H	W	Q

Calculate transition probabilities



Number of sequences in MSA with a gap in M5 and starting insertion(s) after M5

Total number of sequences in MSA with a gap in M5

Stockholms universitet

Profile Hidden Markov Model

Constructing (training) HMM profile

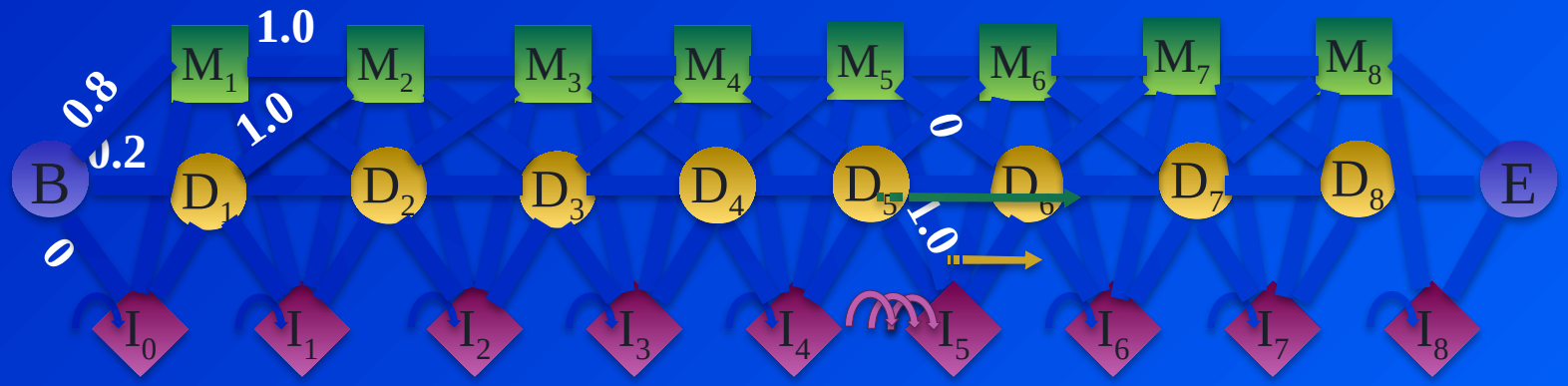
Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	Q	.	H	W	Q

Calculate transition probabilities

Total number of transitions from I_5

Number of inserted residues



Number of sequences in MSA with a residue in M6 and insertion(s) before M6

Number of sequences in MSA with a gap in M6 and insertion(s) before M6

Stockholms universitet

Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Protein 4 **Q N A**

Calculate transition probabilities



Number of residues inserted in MSA after M5

Total number of non-zero transitions from I5

Stockholms universitet

Profile Hidden Markov Model

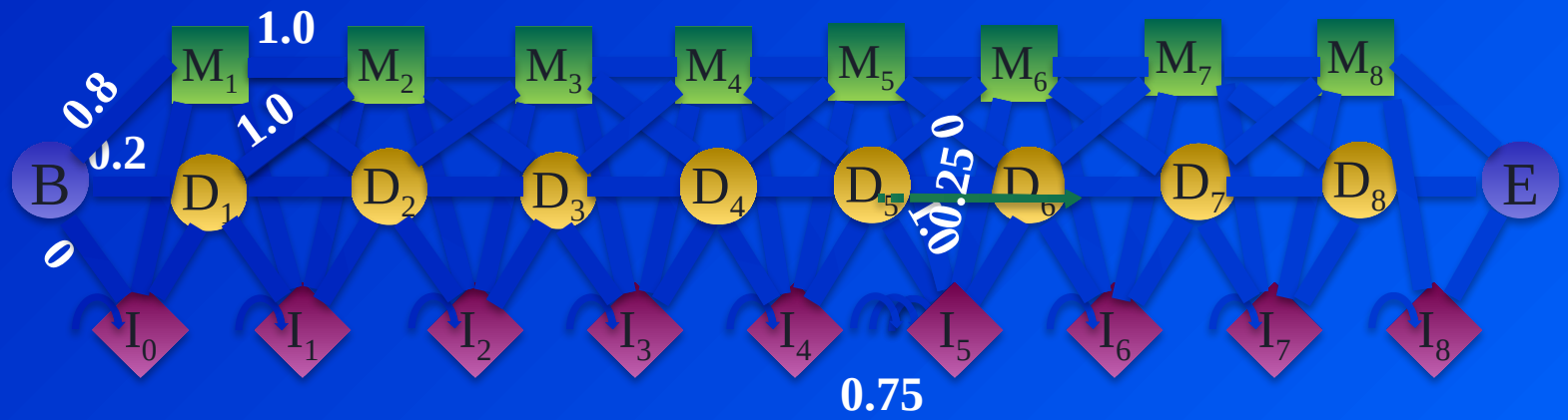
Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	N	W	Q

Protein 4 **A H**

Calculate transition probabilities



Number of sequences with insertion before M6 and a residue in M6

Total number of non-zero transitions from I5

Stockholms universitet

Profile Hidden Markov Model

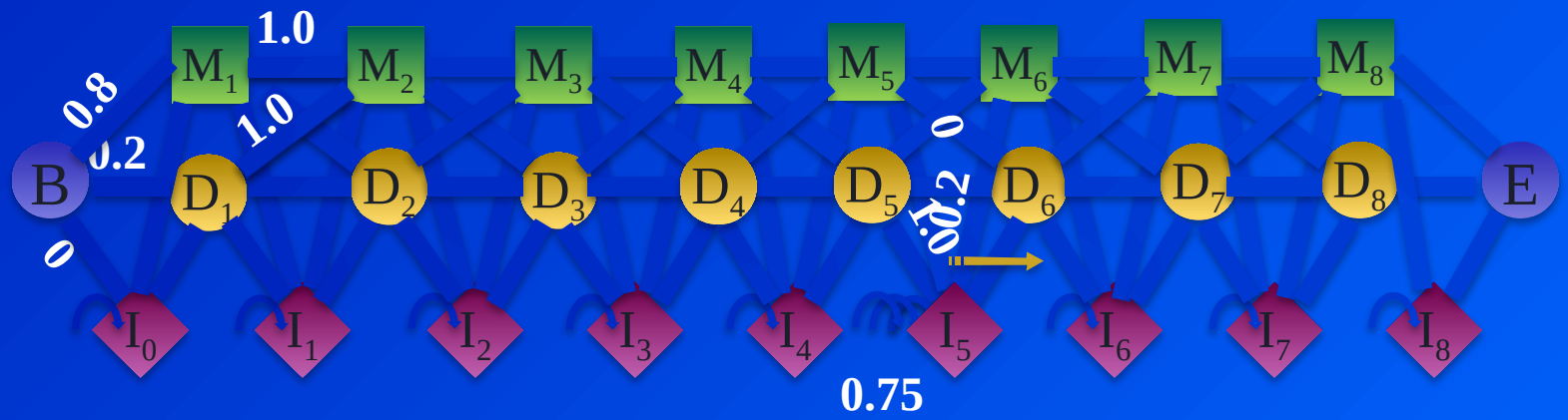
Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	N	W	Q

Protein 4 **A H**

Calculate transition probabilities



Number of sequences with insertion before M6 and a residue in M6

Total number of non-zero transitions from I5

Stockholms universitet

Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

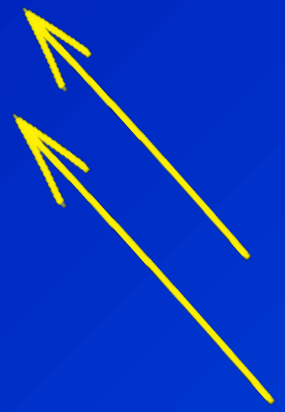
Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

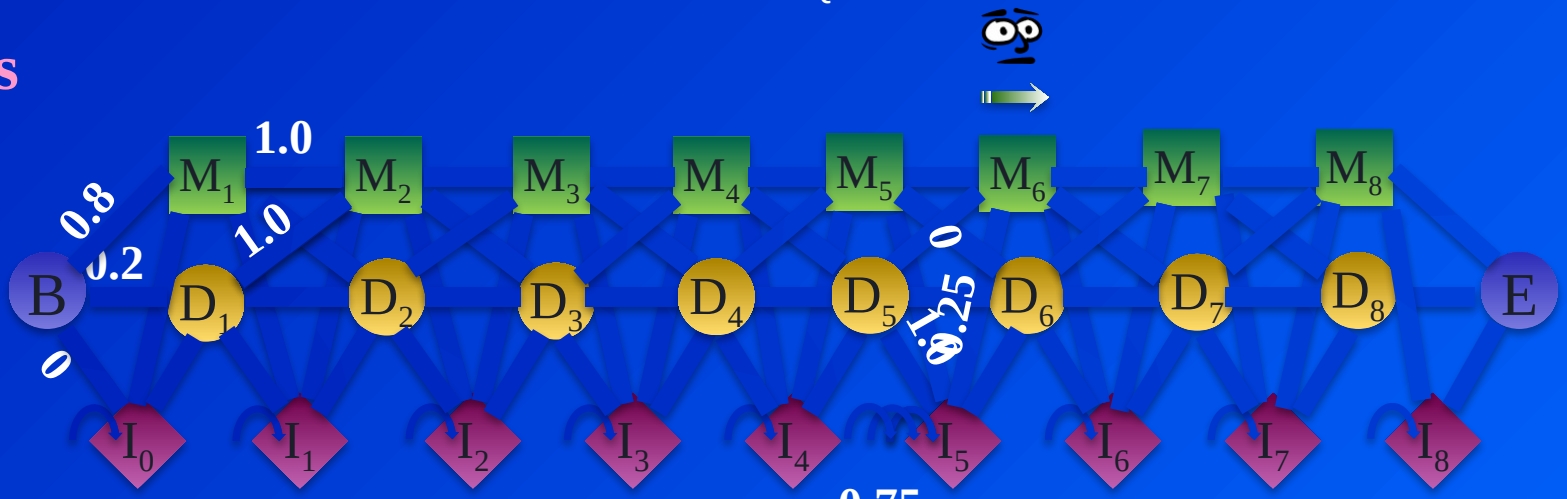
9
K

Protein 4

Calculate emission probabilities

From M_6





Number of Lys residues in MSA column M6

Total number of residues in MSA column M6

Profile Hidden Markov Model

Stockholms universitet

Constructing (training) HMM profile

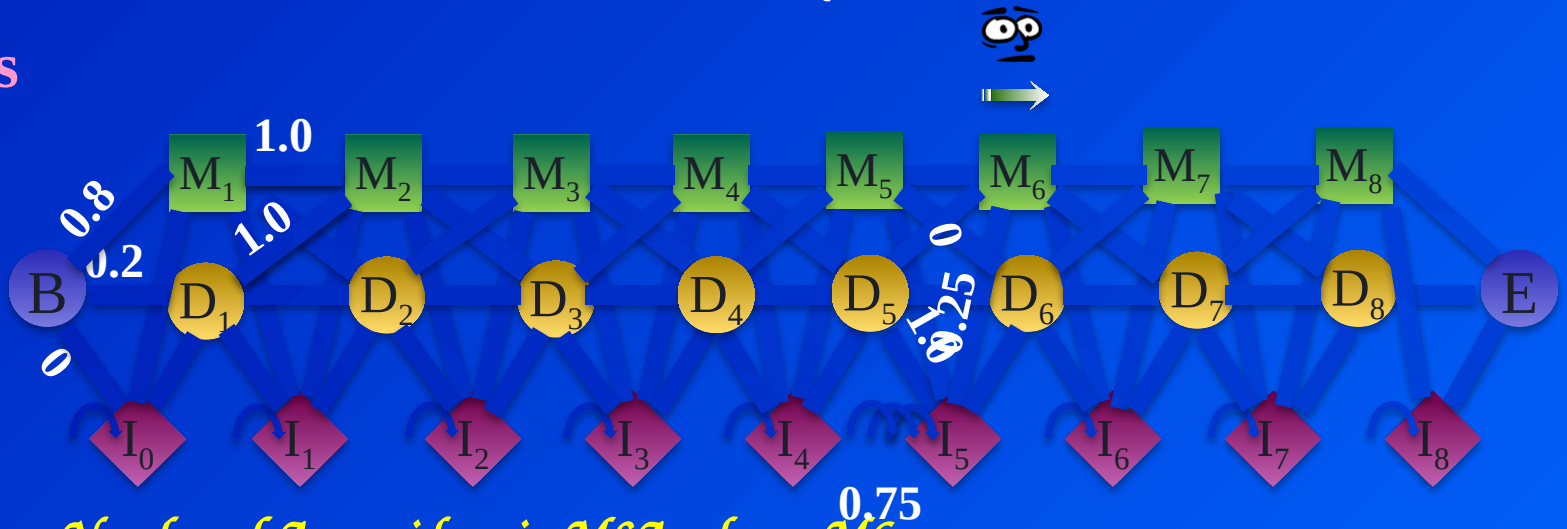
Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	D	-
Protein 3	-	G	K	S	R	.	.	.	R	I	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

9
R
R
Protein 4

Calculate emission probabilities

From M_6



Number of Arg residues in MSA column M6

Total number of residues in MSA column M6

Profile Hidden Markov Model

Constructing (training) HMM profile

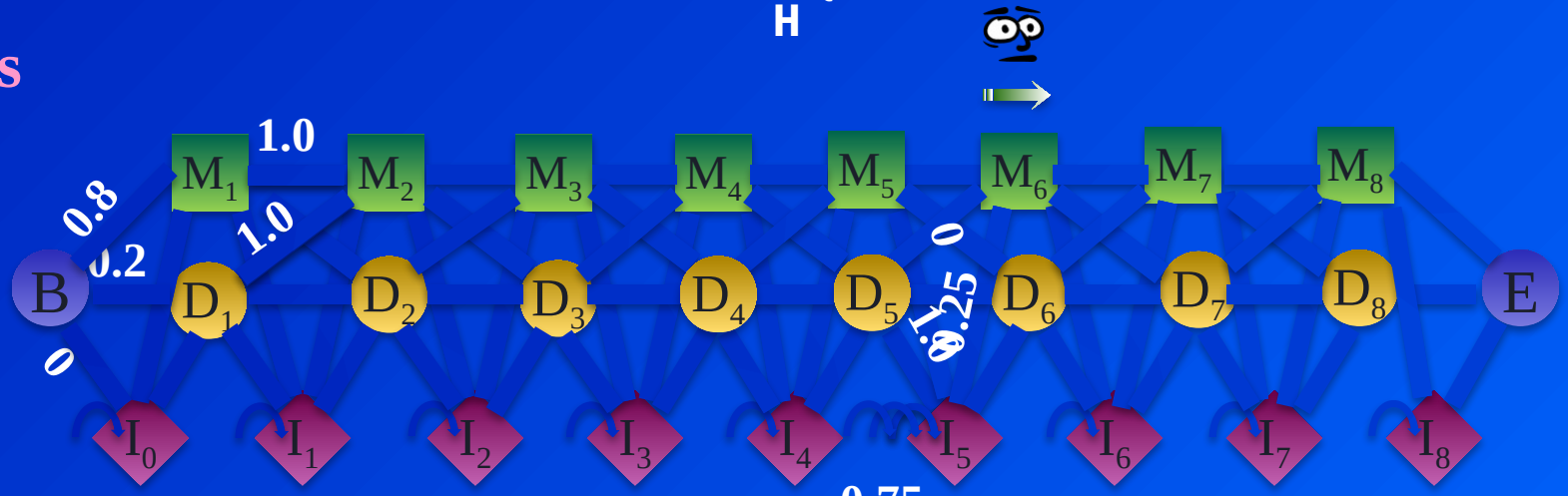
Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

9
H
H

Calculate emission probabilities

From M_6



Number of His residues in MSA column M6

Total number of residues in MSA column M6

Profile Hidden Markov Model

Stockholms universitet

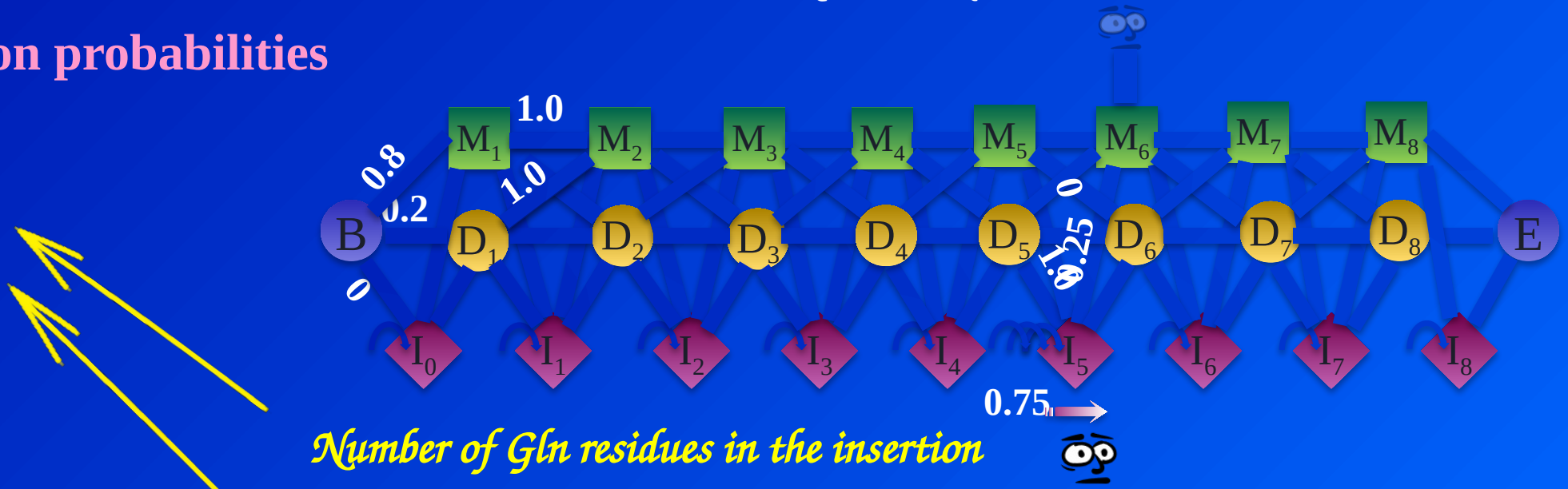
Constructing (training) HMM profile

Multiple sequence Alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	W	Q	

Calculate emission probabilities

From I_5



Number of Gln residues in the insertion

Total number of residues in the insertion

Profile Hidden Markov Model

Stockholms universitet

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	W	Q	

Add pseudocounts

For transition probabilities





Stockholms universitet

Profile Hidden Markov Model

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	A	W	Q

Add pseudocounts

For transition probabilities





Profile Hidden Markov Model

Stockholms universitet

Constructing (training) HMM profile

Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	W	Q	

Protein 4 Q N A

Add pseudocounts

For transition probabilities





Profile Hidden Markov Model

Stockholms universitet

Constructing (training) HMM profile

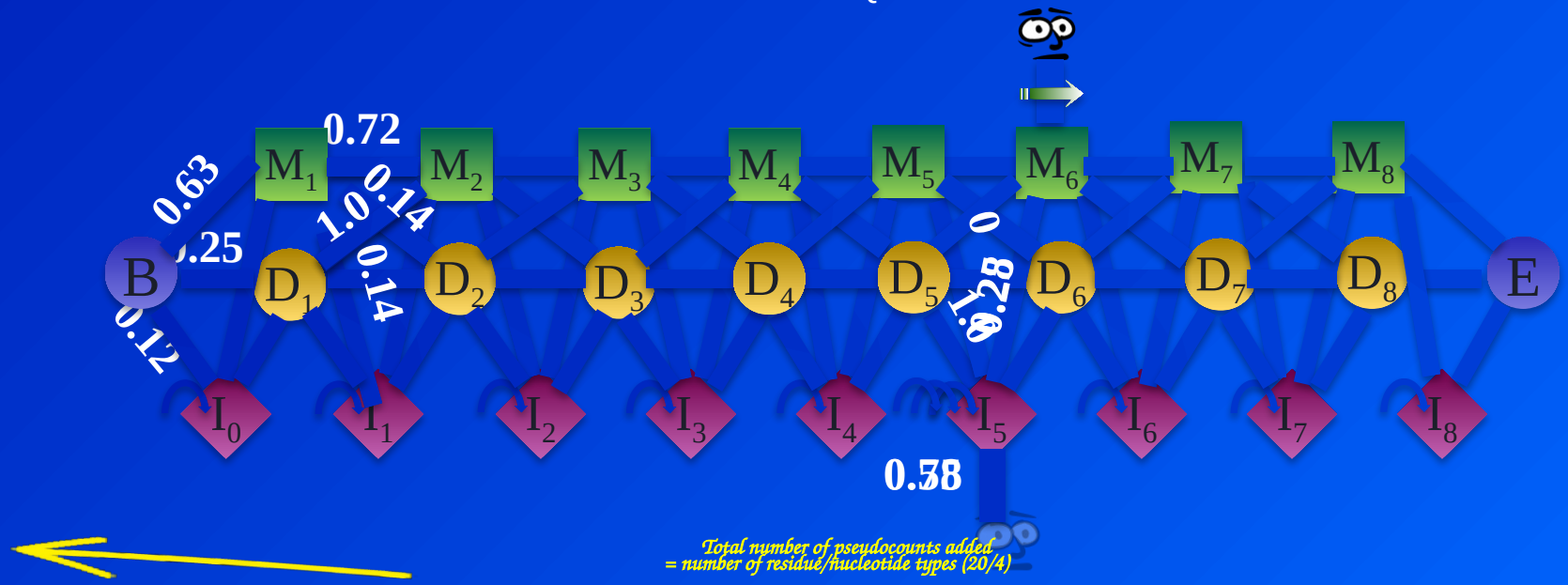
Multiple sequence alignment

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Add pseudocounts

For emission probabilities





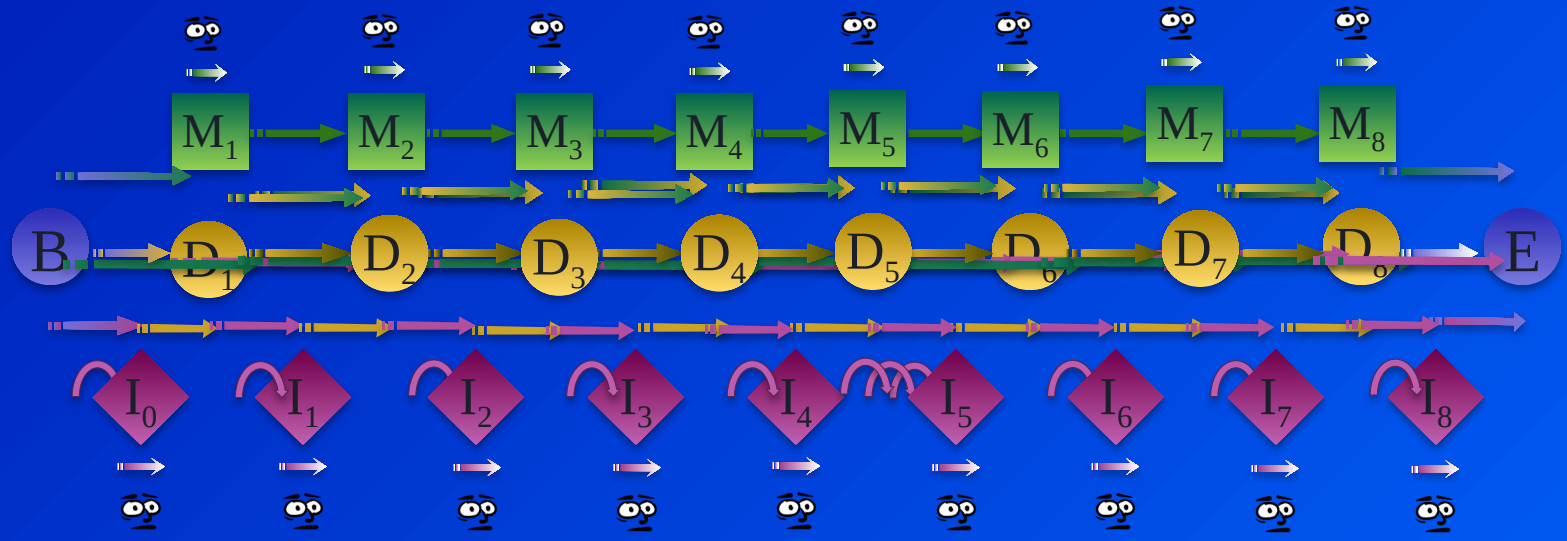
Total number of pseudocounts added = number of residue/nucleotide types (20/4)

Stockholms universitet

Profile Hidden Markov Model

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Using HMM profile

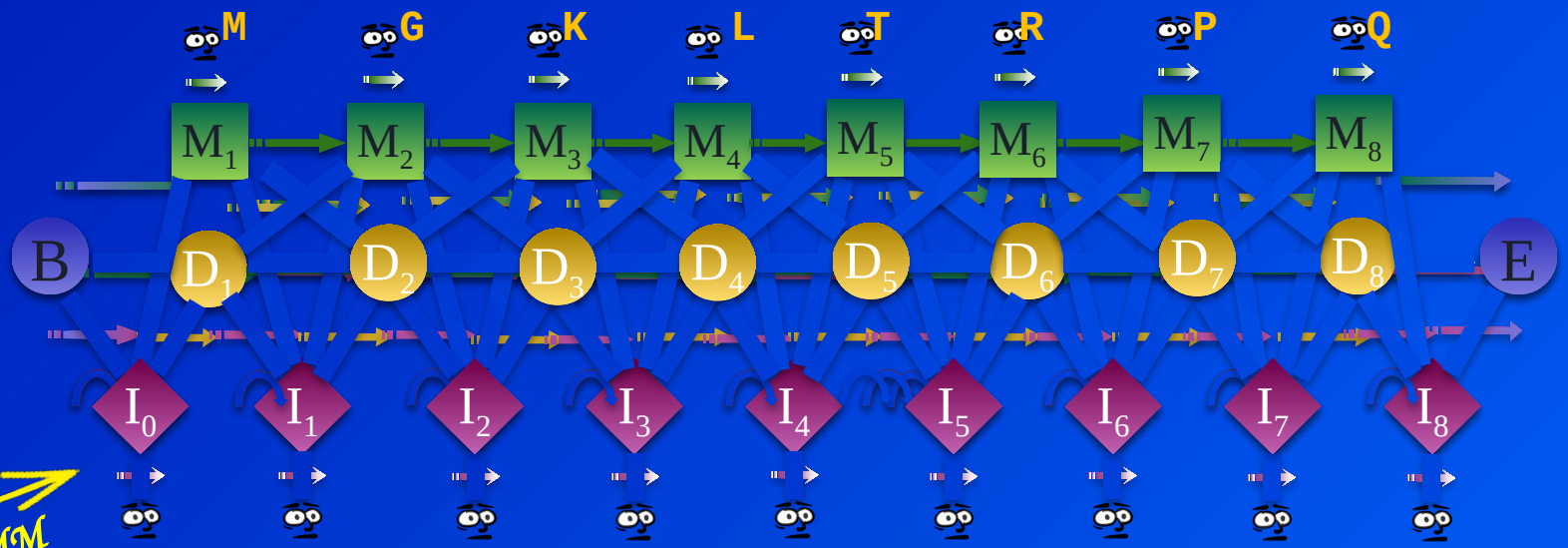


Stockholms universitet

Profile Hidden Markov Model

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Using HMM profile



Scoring an aligned sequence

M G K L T - - - R P Q

Path in HMM

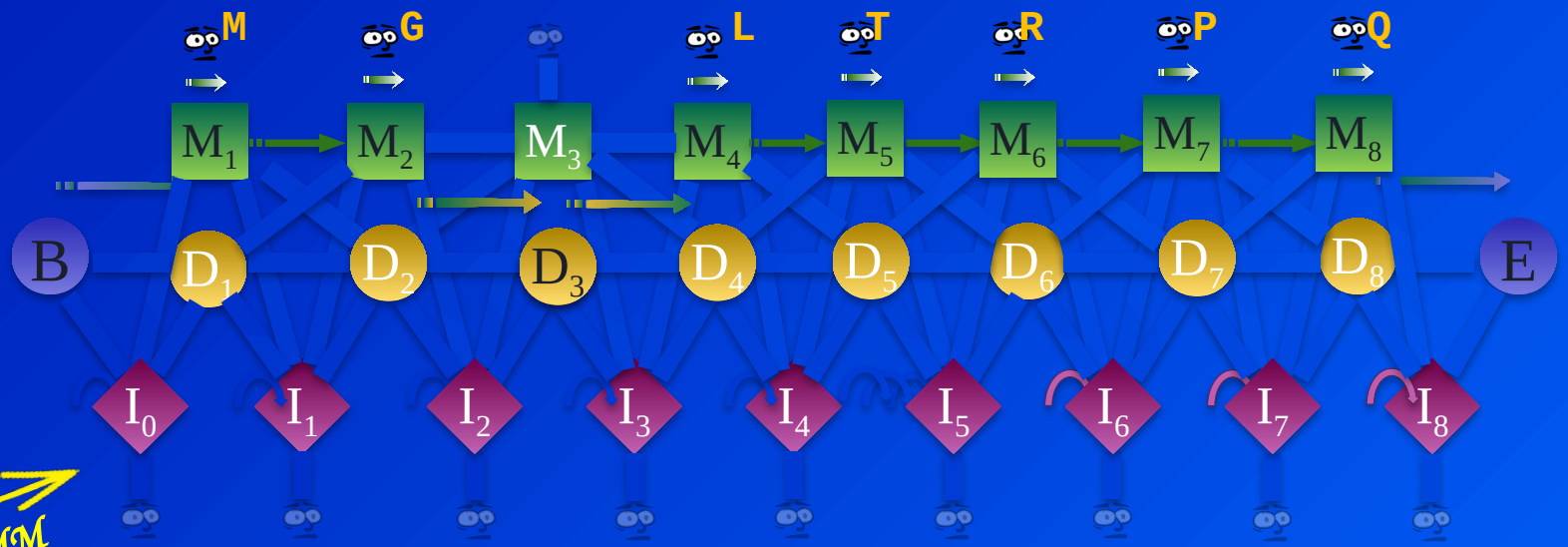
Score

Stockholms universitet

Profile Hidden Markov Model

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Using HMM profile



Scoring an aligned sequence

M G - L T - - - R P Q

Path in HMM

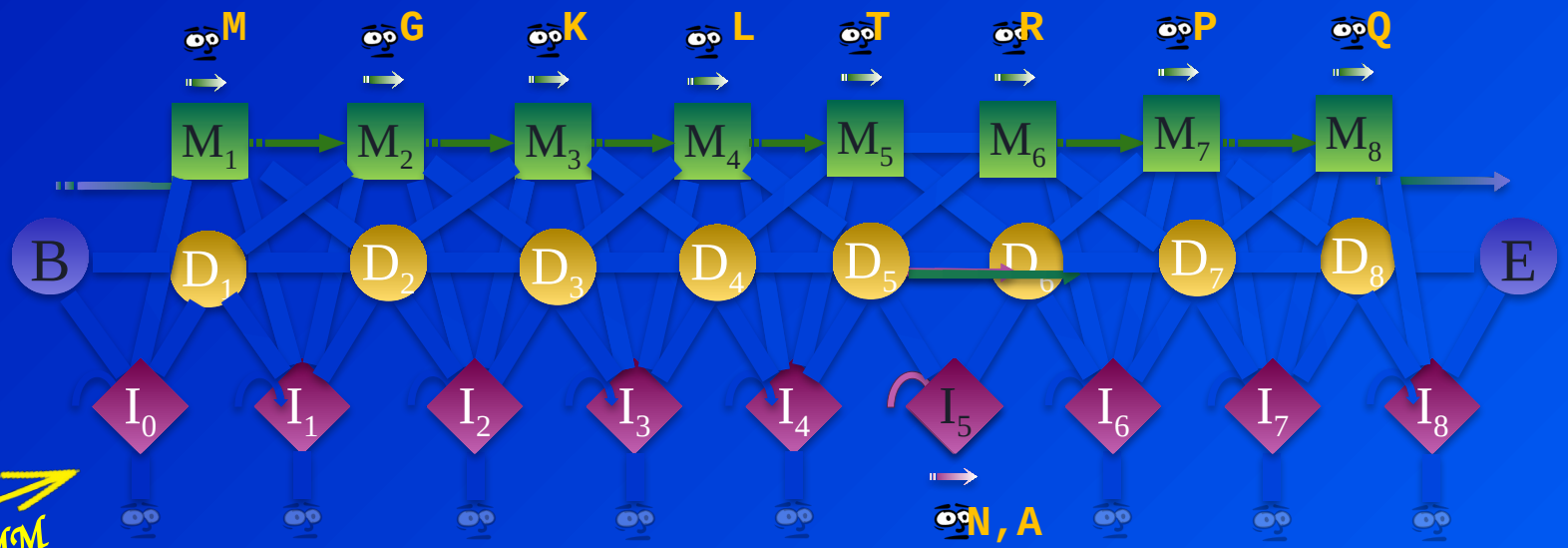
Score

Stockholms universitet

Profile Hidden Markov Model

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q

Using HMM profile



Scoring an aligned sequence

M G K L T - N A R P Q

Path in HMM

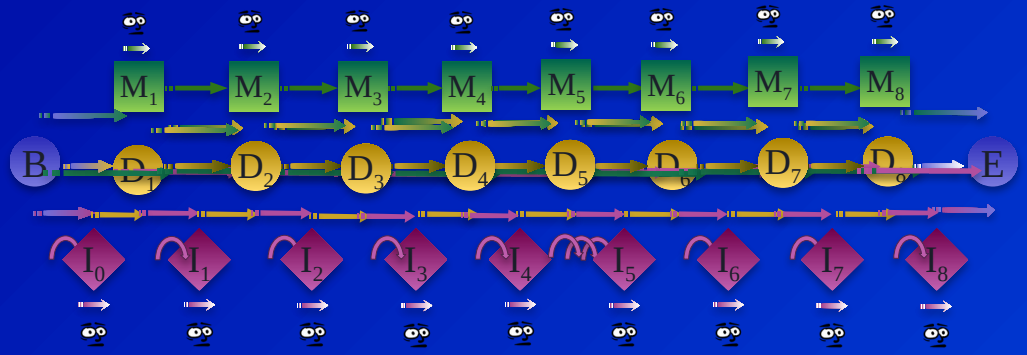
Score

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

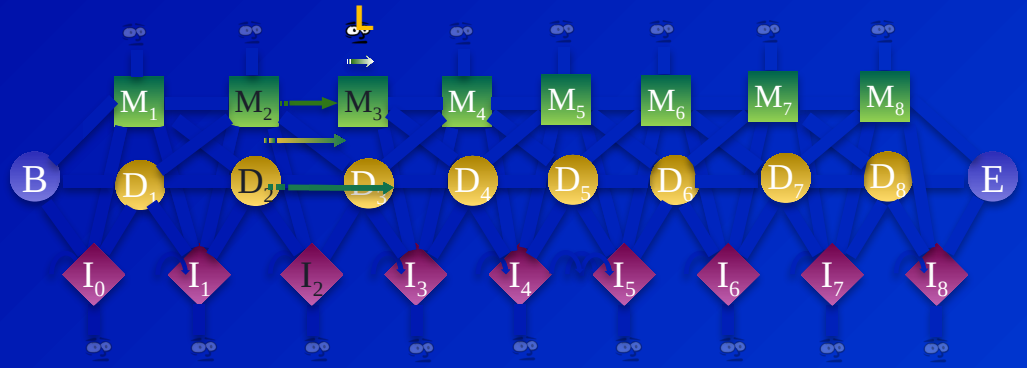
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

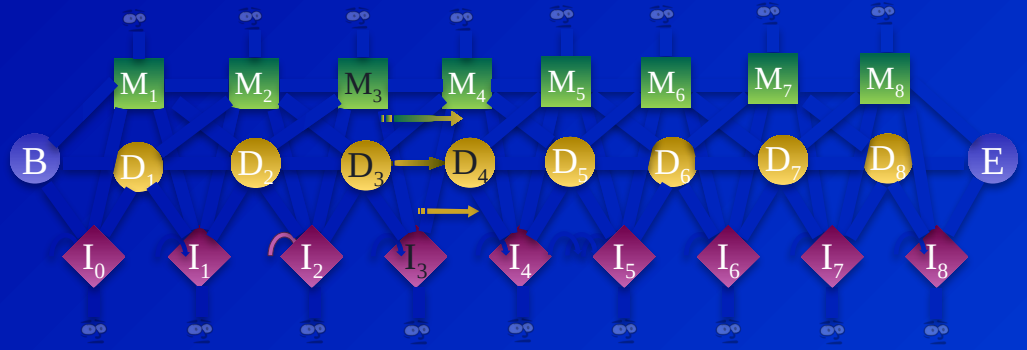
HMM state		Residue					R	P	Q
	(M)	S_{D1} (G)	S_{D1} (L)	S_{D1} (T			0	0	0
	(M)	S_{I1} (G)	S_{I1} (L)	S_{I1} (T			S_{I0}(R)	S_{I0}(P)	S_{I0}(Q)
I_0	(M)	**S_{M2} (G)**	S_{M2} (L)	S_{M2} (T			S_{M1}(R)	S_{M1}(P)	S_{M1}(Q)
M_1	(M)	S_{D2} (G)	**S_{D2} (L)**	S_{D2} (T			S_{D1}(R)	S_{D1}(P)	S_{D1}(Q)
D_1	(M)	**S_{I2} (G)**	S_{I2} (L)	S_{I2} (T			S_{I1}(R)	S_{I1}(P)	S_{I1}(Q)
I_1	(M)	S_{M3} (G)	**S_{M3} (L)**	S_{M3} (T			S_{M2}(R)	S_{M2}(P)	S_{M2}(Q)
M_2	(M)	S_{D3} (G)	S_{D3} (L)	S_{D3} (T			S_{D2}(R)	S_{D2}(P)	S_{D2}(Q)
D_2	(M)	S_{I3} (G)	S_{I3} (L)	S_{I3} (T			S_{I2}(R)	S_{I2}(P)	S_{I2}(Q)
I_2							S_{M3}(R)	S_{M3}(P)	S_{M3}(Q)
M_3									
D_3	0	S_{D3}(M)	S_{D3}(G)	S_{D3}(L)	S_{D3}(T)	S_{D3}(A)	S_{D3}(R)	S_{D3}(P)	S_{D3}(Q)
I_3	0	S_{I3}(M)	S_{I3}(G)	S_{I3}(L)	S_{I3}(T)	S_{I3}(A)	S_{I3}(R)	S_{I3}(P)	S_{I3}(Q)
M_4	0	S_{M4}(M)	S_{M4}(G)	S_{M4}(L)	S_{M4}(T)	S_{M4}(A)	S_{M4}(R)	S_{M4}(P)	S_{M4}(Q)
D_4	0	S_{D4}(M)	S_{D4}(G)	S_{D4}(L)	S_{D4}(T)	S_{D4}(A)	S_{D4}(R)	S_{D4}(P)	S_{D4}(Q)
I_4	0	S_{I4}(M)	S_{I4}(G)	S_{I4}(L)	S_{I4}(T)	S_{I4}(A)	S_{I4}(R)	S_{I4}(P)	S_{I4}(Q)
M_5	0	S_{M5}(M)	S_{M5}(G)	S_{M5}(L)	S_{M5}(T)	S_{M5}(A)	S_{M5}(R)	S_{M5}(P)	S_{M5}(Q)
D_5	0	S_{D5}(M)	S_{D5}(G)	S_{D5}(L)	S_{D5}(T)	S_{D5}(A)	S_{D5}(R)	S_{D5}(P)	S_{D5}(Q)
I_5	0	S_{I5}(M)	S_{I5}(G)	S_{I5}(L)	S_{I5}(T)	S_{I5}(A)	S_{I5}(R)	S_{I5}(P)	S_{I5}(Q)
M_6	0	S_{M6}(M)	S_{M6}(G)	S_{M6}(L)	S_{M6}(T)	S_{M6}(A)	S_{M6}(R)	S_{M6}(P)	S_{M6}(Q)
D_6	0	S_{D6}(M)	S_{D6}(G)	S_{D6}(L)	S_{D6}(T)	S_{D6}(A)	S_{D6}(R)	S_{D6}(P)	S_{D6}(Q)

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

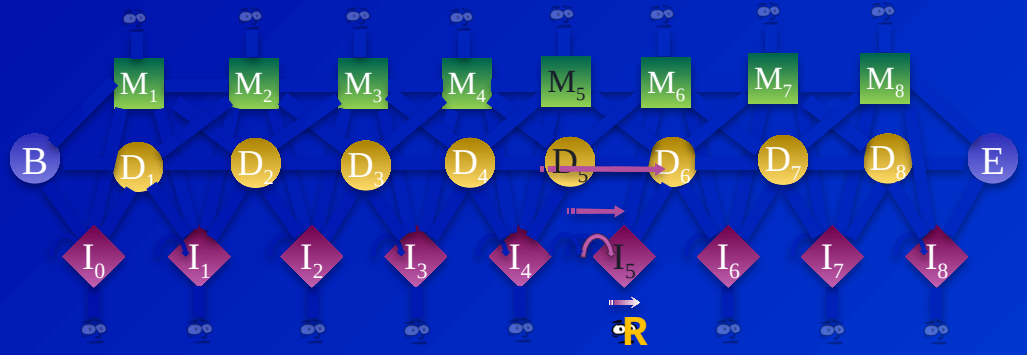
HMM state		Residue							
		S_{D2}(T)	S_{D2}(A)	S_{D2}(R)	S_{D2}(P)				
	0	S_{I2}(T)	S_{I2}(A)	S_{I2}(R)	S_{I2}(P)				
I_0	0	S_{M3}(T)	S_{M3}(A)	S_{M3}(R)	S_{M3}(P)				
M_1	0	S_{D3}(T)	S_{D3}(A)	S_{D3}(R)	S_{D3}(P)				
D_1	0	S_{I3}(T)	S_{I3}(A)	S_{I3}(R)	S_{I3}(P)				
I_1	0	S_{M4}(T)	S_{M4}(A)	S_{M4}(R)	S_{M4}(P)				
M_2	0	S_{D4}(T)	S_{D4}(A)	S_{D4}(R)	S_{D4}(P)				
D_2	0	S_{I4}(T)	S_{I4}(A)	S_{I4}(R)	S_{I4}(P)				
I_2	0	S_{M5}(T)	S_{M5}(A)	S_{M5}(R)	S_{M5}(P)				
M_3	0	S_{M4}(M)	S_{M4}(G)	S_{M4}(L)	S_{M4}(T)	S_{M4}(A)	S_{M4}(R)	S_{M4}(P)	S_{M4}(Q)
D_3	0	S_{D4}(M)	S_{D4}(G)	S_{D4}(L)	S_{D4}(T)	S_{D4}(A)	S_{D4}(R)	S_{D4}(P)	S_{D4}(Q)
I_3	0	S_{I4}(M)	S_{I4}(G)	S_{I4}(L)	S_{I4}(T)	S_{I4}(A)	S_{I4}(R)	S_{I4}(P)	S_{I4}(Q)
M_4	0	S_{M5}(M)	S_{M5}(G)	S_{M5}(L)	S_{M5}(T)	S_{M5}(A)	S_{M5}(R)	S_{M5}(P)	S_{M5}(Q)
D_4	0	S_{D5}(M)	S_{D5}(G)	S_{D5}(L)	S_{D5}(T)	S_{D5}(A)	S_{D5}(R)	S_{D5}(P)	S_{D5}(Q)
I_4	0	S_{I5}(M)	S_{I5}(G)	S_{I5}(L)	S_{I5}(T)	S_{I5}(A)	S_{I5}(R)	S_{I5}(P)	S_{I5}(Q)
M_5	0	S_{M6}(M)	S_{M6}(G)	S_{M6}(L)	S_{M6}(T)	S_{M6}(A)	S_{M6}(R)	S_{M6}(P)	S_{M6}(Q)
D_5	0	S_{D6}(M)	S_{D6}(G)	S_{D6}(L)	S_{D6}(T)	S_{D6}(A)	S_{D6}(R)	S_{D6}(P)	S_{D6}(Q)
I_5	0								

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

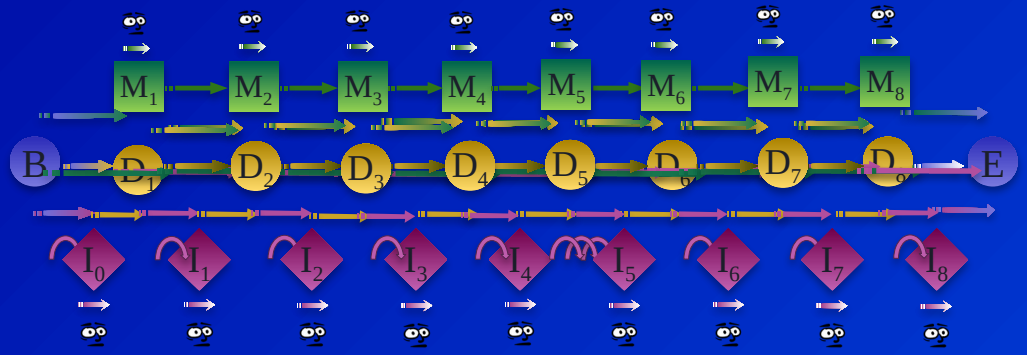
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$					
D_2	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$					
I_2	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$					
M_3	$S_{M5}(T)$	**$S_{M5}(A)$**	$S_{M5}(R)$	$S_{M5}(P)$					
D_3	$S_{D5}(T)$	$S_{D5}(A)$	**$S_{D5}(R)$**	$S_{D5}(P)$					
I_3	$S_{I5}(T)$	**$S_{I5}(A)$**	**$S_{I5}(R)$**	$S_{I5}(P)$					
M_4	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$					
D_4	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$					
I_4	$S(T)$	$S(A)$	$S(R)$	$S(P)$					
M_5									
D_5									
I_5									
M_6									
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$
I_6	0	$S(M)$	$S(G)$	$S(L)$	$S(T)$	$S(A)$	$S(R)$	$S(P)$	$S(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

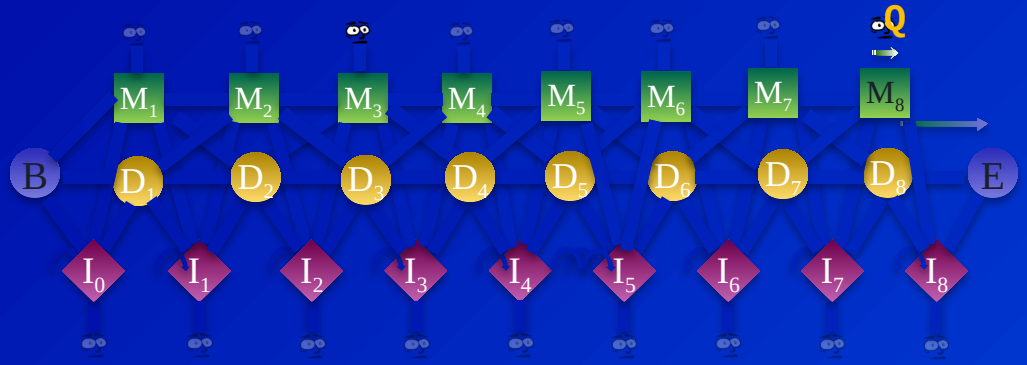
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$
I_6	0	$S_{I6}(M)$	$S_{I6}(G)$	$S_{I6}(L)$	$S_{I6}(T)$	$S_{I6}(A)$	$S_{I6}(R)$	$S_{I6}(P)$	$S_{I6}(Q)$
M_7	0	$S_{M7}(M)$	$S_{M7}(G)$	$S_{M7}(L)$	$S_{M7}(T)$	$S_{M7}(A)$	$S_{M7}(R)$	$S_{M7}(P)$	$S_{M7}(Q)$
D_7	0	$S_{D7}(M)$	$S_{D7}(G)$	$S_{D7}(L)$	$S_{D7}(T)$	$S_{D7}(A)$	$S_{D7}(R)$	$S_{D7}(P)$	$S_{D7}(Q)$
I_7	0	$S_{I7}(M)$	$S_{I7}(G)$	$S_{I7}(L)$	$S_{I7}(T)$	$S_{I7}(A)$	$S_{I7}(R)$	$S_{I7}(P)$	$S_{I7}(Q)$
M_8	0	$S_{M8}(M)$	$S_{M8}(G)$	$S_{M8}(L)$	$S_{M8}(T)$	$S_{M8}(A)$	$S_{M8}(R)$	$S_{M8}(P)$	$S_{M8}(Q)$
D_8	0	$S_{D8}(M)$	$S_{D8}(G)$	$S_{D8}(L)$	$S_{D8}(T)$	$S_{D8}(A)$	$S_{D8}(R)$	$S_{D8}(P)$	$S_{D8}(Q)$
I_8	0	$S_{I8}(M)$	$S_{I8}(G)$	$S_{I8}(L)$	$S_{I8}(T)$	$S_{I8}(A)$	$S_{I8}(R)$	$S_{I8}(P)$	$S_{I8}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: **M G L T A R P Q**

Dynamical programming matrix

Traceback

HMM path: M

Aligned sequence: M

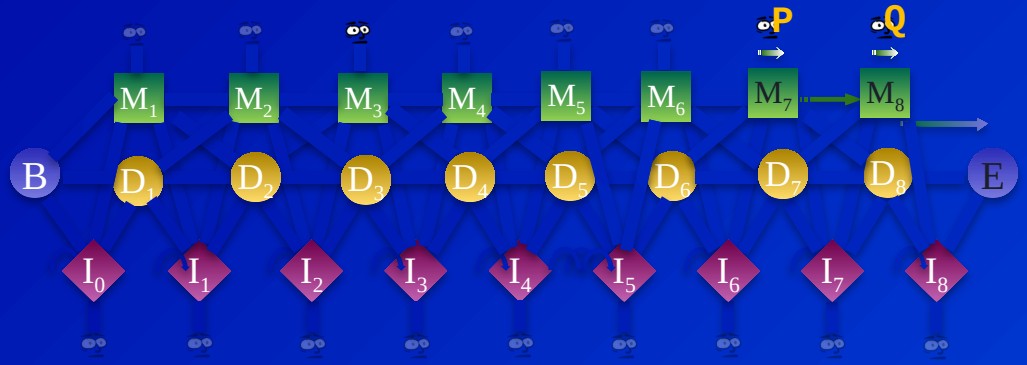
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: M

Aligned sequence: M

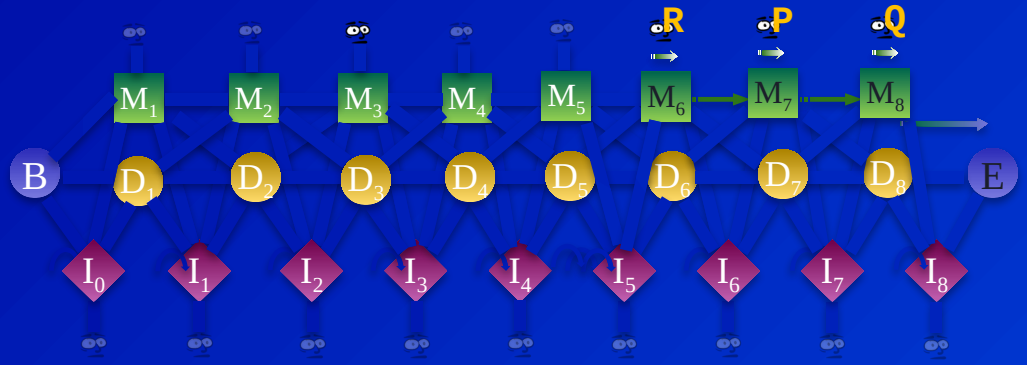
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: [redacted] M8

Aligned sequence: M [redacted] Q

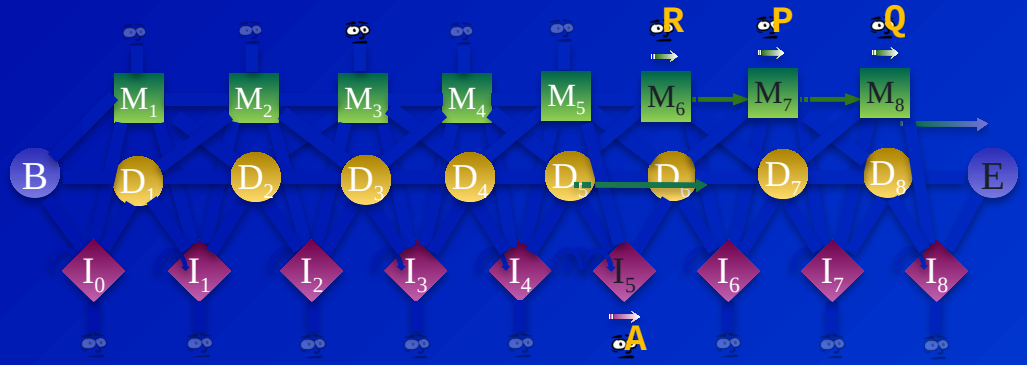
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: [redacted] M7 M8

Aligned sequence: M [redacted] P Q

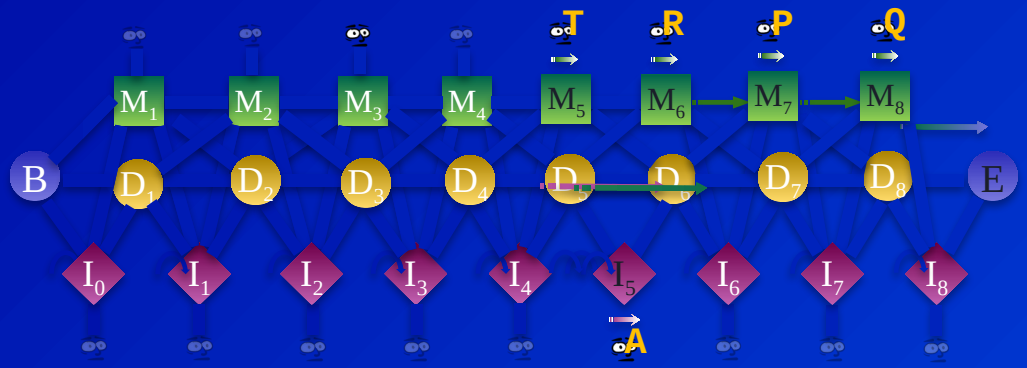
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: M [redacted] 5▫M6▫M7▫M8

Aligned sequence: M [redacted] . A R P Q

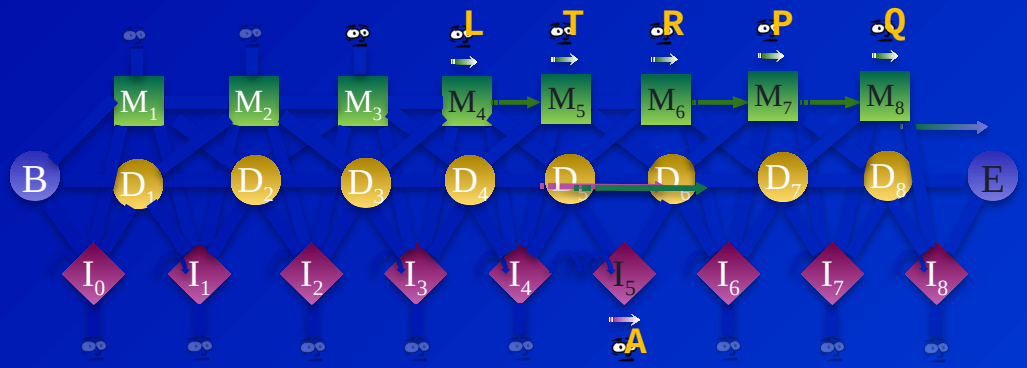
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: M [redacted] M5 I5 M6 M7 M8

Aligned sequence: M [redacted] T . . A R P Q

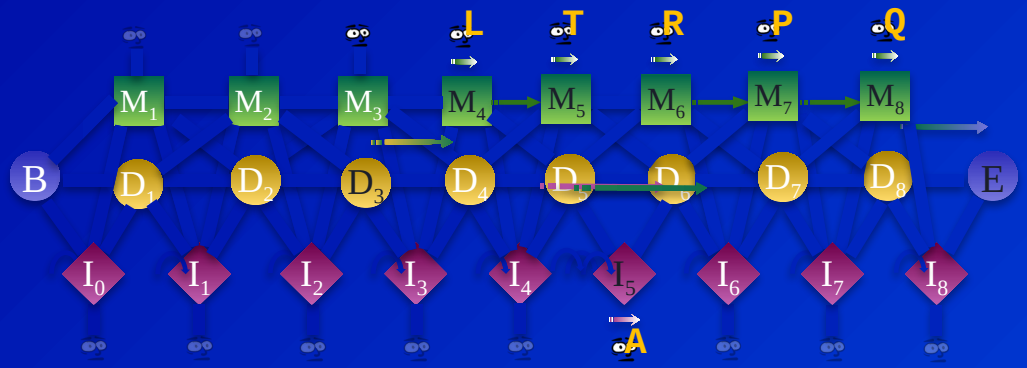
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: [redacted] M4 M5 I5 M6 M7 M8

Aligned sequence: M [redacted] L T . . A R P Q

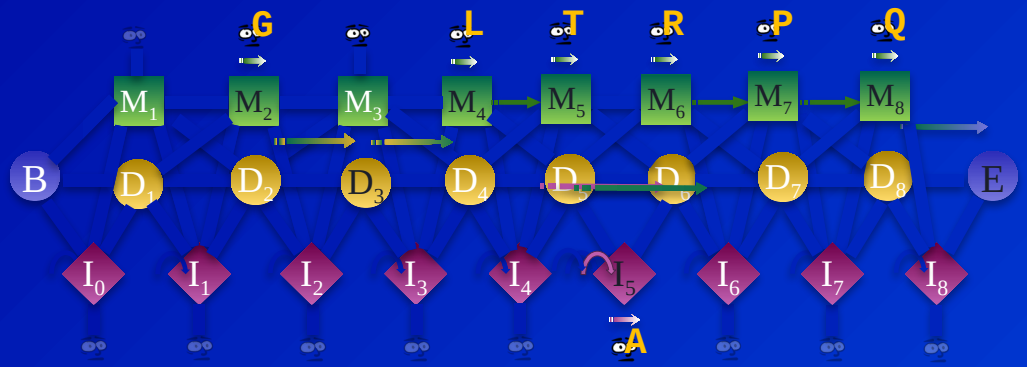
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	**$S_{D3}(G)$**	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: M [redacted] D3 M4 M5 I5 M6 M7 M8

Aligned sequence: M [redacted] - L T . . A R P Q

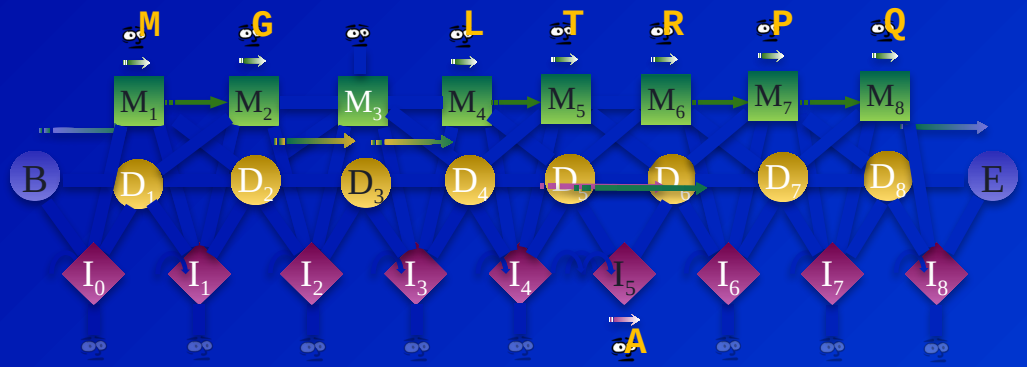
HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	**$S_{M2}(G)$**	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	**$S_{D3}(G)$**	**$S_{D3}(L)$**	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	**$S_{M4}(L)$**	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	**$S_{M5}(T)$**	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	**$S_{I5}(A)$**	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	**$S_{M6}(R)$**	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

Profile Hidden Markov Model

Stockholms universitet

Using HMM profile

Position	1	2	3	4	5	6	7	8	9	10	11
Protein 1	M	A	K	L	T	.	.	.	K	-	D
Protein 2	M	A	K	L	T	.	.	.	R	P	-
Protein 3	-	G	K	S	R	.	.	.	R	T	E
Protein 4	M	A	R	S	-	Q	N	A	H	Y	N
Protein 5	M	A	R	S	R	.	.	.	H	W	Q



Aligning sequence (Vitebri algorithm)

Query sequence: M G L T A R P Q

Dynamical programming matrix

Traceback

HMM path: M1 M2 D3 M4 M5 I5 M6 M7 M8

Aligned sequence: M G - L T . . A R P Q

HMM state		Residue							
		M	G	L	T	A	R	P	Q
	0	0	0	0	0	0	0	0	0
I_0	0	$S_{I0}(M)$	$S_{I0}(G)$	$S_{I0}(L)$	$S_{I0}(T)$	$S_{I0}(A)$	$S_{I0}(R)$	$S_{I0}(P)$	$S_{I0}(Q)$
M_1	0	$S_{M1}(M)$	$S_{M1}(G)$	$S_{M1}(L)$	$S_{M1}(T)$	$S_{M1}(A)$	$S_{M1}(R)$	$S_{M1}(P)$	$S_{M1}(Q)$
D_1	0	$S_{D1}(M)$	$S_{D1}(G)$	$S_{D1}(L)$	$S_{D1}(T)$	$S_{D1}(A)$	$S_{D1}(R)$	$S_{D1}(P)$	$S_{D1}(Q)$
I_1	0	$S_{I1}(M)$	$S_{I1}(G)$	$S_{I1}(L)$	$S_{I1}(T)$	$S_{I1}(A)$	$S_{I1}(R)$	$S_{I1}(P)$	$S_{I1}(Q)$
M_2	0	$S_{M2}(M)$	$S_{M2}(G)$	$S_{M2}(L)$	$S_{M2}(T)$	$S_{M2}(A)$	$S_{M2}(R)$	$S_{M2}(P)$	$S_{M2}(Q)$
D_2	0	$S_{D2}(M)$	$S_{D2}(G)$	$S_{D2}(L)$	$S_{D2}(T)$	$S_{D2}(A)$	$S_{D2}(R)$	$S_{D2}(P)$	$S_{D2}(Q)$
I_2	0	$S_{I2}(M)$	$S_{I2}(G)$	$S_{I2}(L)$	$S_{I2}(T)$	$S_{I2}(A)$	$S_{I2}(R)$	$S_{I2}(P)$	$S_{I2}(Q)$
M_3	0	$S_{M3}(M)$	$S_{M3}(G)$	$S_{M3}(L)$	$S_{M3}(T)$	$S_{M3}(A)$	$S_{M3}(R)$	$S_{M3}(P)$	$S_{M3}(Q)$
D_3	0	$S_{D3}(M)$	$S_{D3}(G)$	$S_{D3}(L)$	$S_{D3}(T)$	$S_{D3}(A)$	$S_{D3}(R)$	$S_{D3}(P)$	$S_{D3}(Q)$
I_3	0	$S_{I3}(M)$	$S_{I3}(G)$	$S_{I3}(L)$	$S_{I3}(T)$	$S_{I3}(A)$	$S_{I3}(R)$	$S_{I3}(P)$	$S_{I3}(Q)$
M_4	0	$S_{M4}(M)$	$S_{M4}(G)$	$S_{M4}(L)$	$S_{M4}(T)$	$S_{M4}(A)$	$S_{M4}(R)$	$S_{M4}(P)$	$S_{M4}(Q)$
D_4	0	$S_{D4}(M)$	$S_{D4}(G)$	$S_{D4}(L)$	$S_{D4}(T)$	$S_{D4}(A)$	$S_{D4}(R)$	$S_{D4}(P)$	$S_{D4}(Q)$
I_4	0	$S_{I4}(M)$	$S_{I4}(G)$	$S_{I4}(L)$	$S_{I4}(T)$	$S_{I4}(A)$	$S_{I4}(R)$	$S_{I4}(P)$	$S_{I4}(Q)$
M_5	0	$S_{M5}(M)$	$S_{M5}(G)$	$S_{M5}(L)$	$S_{M5}(T)$	$S_{M5}(A)$	$S_{M5}(R)$	$S_{M5}(P)$	$S_{M5}(Q)$
D_5	0	$S_{D5}(M)$	$S_{D5}(G)$	$S_{D5}(L)$	$S_{D5}(T)$	$S_{D5}(A)$	$S_{D5}(R)$	$S_{D5}(P)$	$S_{D5}(Q)$
I_5	0	$S_{I5}(M)$	$S_{I5}(G)$	$S_{I5}(L)$	$S_{I5}(T)$	$S_{I5}(A)$	$S_{I5}(R)$	$S_{I5}(P)$	$S_{I5}(Q)$
M_6	0	$S_{M6}(M)$	$S_{M6}(G)$	$S_{M6}(L)$	$S_{M6}(T)$	$S_{M6}(A)$	$S_{M6}(R)$	$S_{M6}(P)$	$S_{M6}(Q)$
D_6	0	$S_{D6}(M)$	$S_{D6}(G)$	$S_{D6}(L)$	$S_{D6}(T)$	$S_{D6}(A)$	$S_{D6}(R)$	$S_{D6}(P)$	$S_{D6}(Q)$

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Some concluding remarks

Word search

Sequence profile search

Hidden Markov model